# FedOCR: Outlier Client Removal in Federated Learning


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***Abstract*—Federated learning keeps data on devices, but noisy labels and non-IID data can still hurt training. Many existing defenses clean data during local training at the sample level, which can improve accuracy but often adds extra computation, communication, and privacy-sensitive bookkeeping.**

**We propose FedOCR, a simple pre-training filter that removes risky clients before federated optimization starts. FedOCR scores each client in two ways: it checks whether the client looks reliable on a small probing set, and it checks whether the remaining clients still cover the target distribution well. Using these signals, FedOCR keeps a client set that is both cleaner and more balanced.**

**Experiments on MedMNIST, CIFAR-10, and CIFAR-100 show that FedOCR improves robustness under noisy and non-IID settings, reduces the cost of reaching a target accuracy, and can work together with training-stage noisy-label correction. Overall, FedOCR provides a practical front-end step for robust federated learning.**



***Index Terms*—Federated learning, client selection, label noise, Non-IID data, robust learning**


## I. Introduction

Federated learning (FL) preserves data privacy by keeping data decentralized, yet its practical adoption is hindered by substantial communication and computation overhead. Frequent transmission of large model updates and the limited computational capacity of edge devices are widely recognized as major barriers to real-world deployment [1]–[5]. This challenge becomes particularly severe in scenarios with heterogeneous client data quality, where label noise and non-IID distributions coexist and often require additional auditing costs per round [6]–[9]. Thus, a central question arises: how can FL maintain robust performance at low cost under such challenging conditions?

To address robustness under noisy labels, numerous FL methods, commonly referred to as noisy-label FL, have been developed. These approaches typically operate at the sample level during training, using techniques such as sample reweighting, label correction or peer-model filtering to mitigate the impact of corrupted examples [10]–[18]. This design is well motivated: label noise occurs at the instance level, and directly handling suspicious samples provides the finest-grained control. From an outlier-processing perspective, these methods can be interpreted as sample-level outlier correction during training.

However, fine-grained correction inevitably incurs non-negligible system costs. When certain clients are persistently noisy or strongly target-mismatched, admitting them in every round is inefficient: the system still spends communication and local computation on these risk-prone clients before their harmful samples are filtered, reweighted, or corrected [5], [15], [18]. FedENLC further makes this burden explicit, showing that label correction can require substantially more computation than standard training [19]. Therefore, sample-level processing alone cannot fully address the efficiency requirement of robust FL.

To achieve both robustness and efficiency, we shift the outlier-removal unit from individual samples to client-side data blocks and perform **outlier-client removal at the pre-training stage**. This viewpoint is natural in FL: although noise occurs at the sample level, communication, local training, and aggregation are all organized at the client level. Leveraging the block-structured nature of federated data, client-level removal can be equivalently formulated as block-constrained client-level outlier removal (BC-ORS; see definition 2).

Our key idea is to exclude high-risk client blocks once before formal federated optimization begins. Because this decision is made at a coarse granularity, it can reduce communication and computation costs throughout subsequent training rounds. Beyond cost savings, client-level outlier removal brings three additional advantages. First, it matches the operational granularity of FL, where the server can admit or reject clients but cannot directly manipulate private samples. Second, it is privacy-compatible, as it does not require the server to request or exchange additional sample-level information [5].

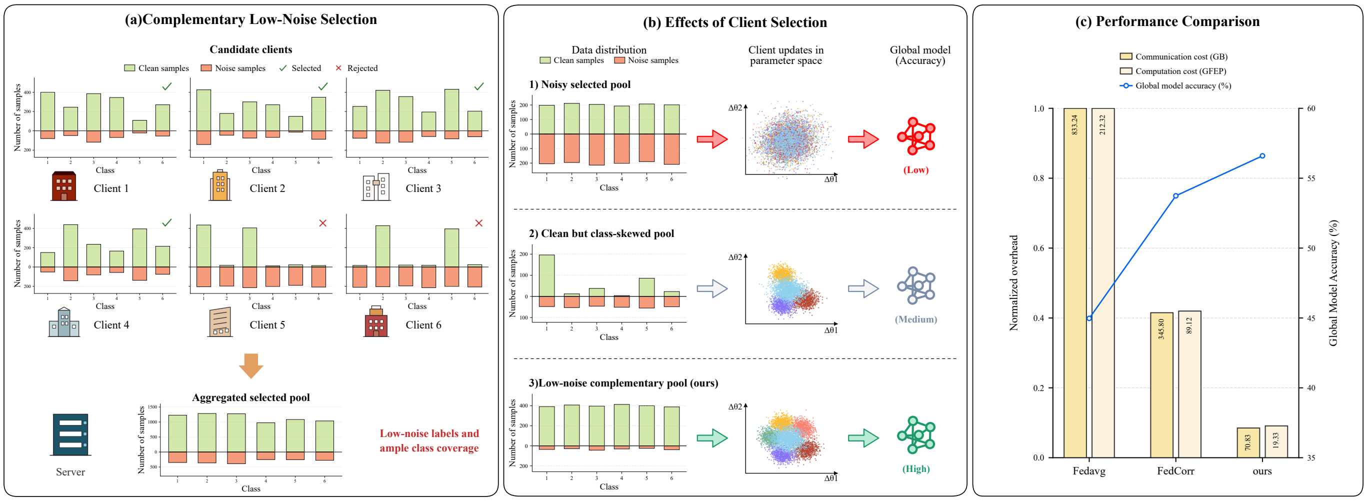


Fig. 1. Motivation and effect of FedOCR under heterogeneous noisy-label federated learning. FedOCR performs client-block-level outlier removal before training, retaining a low-risk and complementary participant pool that reduces corrupted updates and improves training efficiency.

Third, it provides robust pre-filtering: by removing risk-prone clients before training, the remaining client pool becomes more reliable, reducing the chance that downstream correction methods are biased by severely corrupted or target-mismatched updates [15], [16], [20].

Based on this principle, we propose **FedOCR**, a reliability- and representativeness-aware framework for pre-training client-level outlier removal. FedOCR operationalizes two core questions: which clients are reliable enough to participate, and whether the retained clients still cover the target-relevant distribution. Specifically, each candidate client first performs a short local adaptation, after which the server evaluates the resulting client model on a server-held probing set. Predictive entropy on the probing set is used to estimate client-block reliability, while KL-based distribution matching evaluates whether the retained client set remains representative of the target-relevant distribution [6], [8], [13], [21]–[24].

FedOCR then uses an adaptive greedy rule to integrate reliability and representativeness signals and construct the retained client set. The key point is that the best set size is not fixed blindly: retaining too few clients may lose target coverage, whereas retaining too many can reintroduce accumulated noise. Accordingly, FedOCR aims to retain a client set that contains sufficient clean data while preserving balanced class coverage. As shown in Fig. 1(b), such a low-noise and complementary client pool yields the best empirical performance.

Since FedOCR operates before the main FL training stage, it can be flexibly stacked with downstream noisy-label FL methods. Rather than replacing sample-level correction, FedOCR improves the admitted client pool before optimization begins, allowing subsequent robust training methods to operate on a cleaner and more reliable set of clients. Our integration experiments confirm this benefit: combining FedOCR with training-stage methods, including FedCor [25], FedCorr [15], and FedNoRo [16], improves both accuracy and efficiency compared with using each method alone. Therefore, when higher final accuracy is desired, FedOCR can be coupled with stronger training-stage noisy-label FL methods.

Our main contributions are summarized as follows:

- We study robust federated learning from a **client-block-level outlier-removal perspective**, shifting the denoising unit from individual noisy samples to locality-constrained client data blocks.
- We propose a **one-shot pre-training client admission stage** that aligns with the operational structure of FL: the server can admit or reject client blocks without accessing private samples, enabling system-aware risk control before formal optimization.
- We formulate a **principled construction of an outlier-removal client set** that jointly accounts for reliability outliers and distributional risks while preserving coverage of target-relevant data.
- We introduce **FedOCR**, which integrates probing-set predictive entropy, KL-based distribution matching, and an adaptive greedy selection rule to build the retained client pool. Experiments validate the design, demonstrating lower communication and computation costs at matched target utility and compatibility with training-stage noisy-label FL methods.

## II. Related Work

*a) **Centralized outlier removal:*** Outlier removal in centralized learning has long focused on instance-level noisy-label filtering and robust training. Classical approaches correct labels, reweight suspicious samples, or identify clean examples by low loss or high confidence, including loss correction, co-teaching, confident learning, and related noisy-label routines [11]–[13], [26], [27]. These methods are effective because the learner can directly inspect all samples and, when available, use validation evidence to identify harmful instances. FedOCR inherits this robustness intuition, but moves the removal decision one level up, to client blocks rather than individual samples.

*b) **Noisy-label learning in FL:*** Federated learning inherits the same noisy-label problem, but the privacy-preserving and communication-constrained setting makes sample-level correction harder to apply repeatedly [9], [28]–[30]. Existing FL methods adapt centralized ideas through label correction, sample reweighting, small-loss selection, peer-model filtering, and local data refinement [14]–[18], [20], [31]. While effective, these approaches often add extra local computation, confidence tracking, state maintenance, or privacy-sensitive signals, and they typically operate after clients have already been admitted to training [5], [15], [18]. FedOCR targets an earlier stage: it screens client blocks before training so that downstream methods can work on a cleaner participant pool.

*c) **Client-level dynamic outlier removal:*** A separate line of work studies client selection to improve FL efficiency and convergence under communication and resource constraints [1], [3], [4], [32]. Representative methods choose clients by estimated utility, loss, gradient correlation, diversity, or cluster-level informativeness [24], [25], [33]–[35]. Most of these methods are round-wise schedulers: they ask which clients are useful for the current communication round, not whether a client should remain in the pool at all. FedOCR extends this client-level pruning view by jointly considering reliability and subset-level representativeness. Because the server cannot inspect raw data directly, these criteria are realized through privacy-compatible probing signals, namely probing-set predictive entropy and KL-based probing distribution matching.

## III. Theoretical Analysis

We study *pre-training outlier removal* for federated learning (FL) from a client-level perspective. Our goal is to select a subset of training samples that minimizes the target risk, while respecting FL constraints that isolate samples into client-specific blocks.

### A. Problem Setup

Let $P_T$ denote the target distribution, and $h \in \mathcal{H}$ any hypothesis with target risk

$$\epsilon_T(h) = \mathbb{E}_{(x,y)\sim P_T}[\ell(h(x), y)]. \tag{1}$$

**Assumption 1** (Existence of risk-prone data)**.** *The candidate training collection $D$ contains a non-empty subset of risk-prone samples $D_{\text{risk}} \subset D$. Samples in $D_{\text{risk}}$ may contain persistent risks, such as corrupted labels or target-mismatched examples, which can harm the target model if they are retained for training.*

**Definition 1** (Sample-level outlier-removal set (ORS))**.** In a centralized setting where all training samples $D$ are accessible, the optimal retained sample subset is

$$S_{\text{ORS}} \in \arg\min_{\emptyset \neq S \subseteq D} \epsilon_T(h_S), \tag{2}$$

where $h_S$ is trained on subset $S$. This defines the *sample-level outlier-removal set*.

**Definition 2** (Block-constrained ORS (BC-ORS) under FL)**.** In FL, samples are partitioned into client blocks $D = \bigsqcup_{i=1}^{N} D_i$, $D_i$ donates private data on client $i$. For a given subset of sample blocks $S^\star = \bigsqcup_{i=1}^{M} D_i$, and $M \subseteq \{1, \dots, N\}$, define the retained sample set The *sample-block-constrained ORS* is

$$S^\star_{\text{BC-ORS}} \in \arg\min_{\emptyset \neq S^\star \subseteq \{1,\dots,N\}} \epsilon_T(h_{S^\star}), \tag{3}$$

where $h_{S^\star}$ is trained on the retained samples in $S^\star$. Compared to centralized ORS, the only constraint is that samples cannot be selected individually; instead, entire client blocks are retained or discarded.

*Remark* 1 (ORS versus BC-ORS)*.* ORS is the sample-level oracle benchmark and BC-ORS is the FL-constrained counterpart. Therefore, BC-ORS is generally slightly suboptimal to ORS in target accuracy. We nevertheless prefer BC-ORS in FL because privacy and locality constraints rule out sample-level auditing, and because finding BC-ORS is orders of magnitude more efficient in communication and computation, as illustrated in Fig. 1(c) and Section V.

### B. A Unified Risk View of Client-Block Outlier Removal

We now formalize why pre-training client-level outlier removal is justified. Since FL cannot realize sample-level ORS, the key question is whether a retained client-block set $S$ lowers the target-risk upper bound. In this section, our analysis shows that two factors matter: unreliable clients can inject noisy supervision, and an imbalanced retained pool can drift away from the target distribution [6]–[8].

**Assumption 2** (Bounded loss)**.** *The loss function is bounded:*

$$0 \le \ell(h(x), y) \le 1, \qquad \forall h \in \mathcal{H},\ (x, y) \in \mathcal{X} \times \mathcal{Y}. \tag{4}$$

**Assumption 3** (Label-shift-dominated heterogeneity)**.** *For every clean client distribution $\mu_i$ and the target distribution $P_T$, the class-conditional distribution is shared:*

$$\mu_i(x, y) = p(x \mid y) q_i(y), \qquad P_T(x, y) = p(x \mid y) q_T(y), \tag{5}$$

*where $q_i \in \Delta^{K-1}$ is the clean label prior of client $i$, and $q_T \in \Delta^{K-1}$ is the target label prior.*

Assumption 3 focuses on label-prior heterogeneity, where clients differ mainly in class proportions rather than in class-conditional feature distributions [6], [23]. Thus, the theoretical bound characterizes a label-shift-dominated case of Non-IID FL.

For a selected subset $S$, define its clean mixture label prior as

$$\bar{q}_S = \sum_{i \in S} w_i(S) q_i, \qquad w_i(S) = \frac{n_i}{\sum_{j \in S} n_j}. \tag{6}$$

Under Assumption 3, the clean selected-client mixture can be written as

$$P^{\text{c}}_S(x, y) = p(x \mid y) \bar{q}_S(y). \tag{7}$$

**Assumption 4** (Label-only noise)**.** *For any selected subset $S$, the clean selected-client mixture $P^{\text{c}}_S(X, Y)$ and the* *noisy selected-client mixture $P_S^n(X, \widetilde{Y})$ share the same input marginal:*

$$P_S^c(X) = P_S^n(X). \tag{8}$$

*That is, label noise only corrupts the clean label $Y$ into the observed label $\widetilde{Y}$.*

This label-only corruption model follows the standard noisy-label learning setting, including symmetric label noise used in many robustness studies [9]–[11].

Define the posterior label-noise discrepancy as

$$\xi_{\mathrm{noise}}(S) = \mathbb{E}_{x \sim P_S^X}\left[\sum_{c=1}^{K}\left|P_S^c(Y = c \mid x) - P_S^n(\widetilde{Y} = c \mid x)\right|\right]. \tag{9}$$

**Lemma 1** (Posterior-noise decomposition)**.** *Under Assumptions 2 and 4, for any $h \in \mathcal{H}$ and any selected subset $S$, let $\epsilon_S^c(h)$ and $\epsilon_S^n(h)$ denote the clean and noisy risks on $S$, respectively. Then,*

$$|\epsilon_S^c(h) - \epsilon_S^n(h)| \le \xi_{\mathrm{noise}}(S). \tag{10}$$

*Consequently,*

$$\epsilon_S^c(h) \le \epsilon_S^n(h) + \xi_{\mathrm{noise}}(S). \tag{11}$$

This lemma should be interpreted as a risk decomposition showing how label corruption enters the target-risk bound. It does not claim that the posterior discrepancy in Eq. (9) is directly observable by the server.

**Lemma 2** (Clean target-transfer under label shift)**.** *Under Assumptions 2 and 3, for any $h \in \mathcal{H}$ and any selected subset $S$,*

$$\epsilon_T(h) \le \epsilon_S^c(h) + \frac{1}{2}\|\bar{q}_S - q_T\|_1. \tag{12}$$

**Theorem 1** (Unified client-block risk bound)**.** *Combining Lemma 2 with Lemma 1, we obtain that, for any selected subset $S$ and any $h \in \mathcal{H}$,*

$$\epsilon_T(h) \le \epsilon_S^n(h) + \xi_{\mathrm{noise}}(S) + \frac{1}{2}\|\bar{q}_S - q_T\|_1. \tag{13}$$

Theorem 1 identifies two population-level factors that should be controlled by a retained client-block set: the noise-contamination term $\xi_{\mathrm{noise}}(S)$ and the distribution-mismatch term $\|\bar{q}_S - q_T\|_1$. Thus, BC-ORS is not merely a heuristic pre-processing step; changing the retained block set $S$ directly changes the upper bound on the target risk, consistent with domain-adaptation views that link target risk to source risk and distribution discrepancy [22], [36]. At the same time, these population quantities are not directly visible to the server before training. The next subsection therefore turns the bound into an operational probing-consistency statement: FedOCR optimizes observable probing-set surrogates, while the approximation gap between these surrogates and the population quantities is kept explicit through residual terms.

### *C. From Population Bound to probing Outlier-Removal Surrogates*

The bound in Theorem 1 lives at the population level, but quantities in bound cannot be inspected by the server under the privacy and locality constraints of FL [5]. FedOCR therefore instantiates the two risk factors through probing-set probing statistics.This step is not treated as an exact substitution. Instead, we formulate a residualized probing bound that separates the observable entropy–KL surrogate from the residual terms induced by probing mismatch. The resulting theory gives a conditional but explicit bridge: when the probing set is target-relevant and the short-adapted client models provide informative uncertainty and class-coverage signatures, optimizing the probing objective is aligned with reducing the population risk factors identified above.

First, we use predictive entropy as a lightweight probing objective for client-block reliability. Figure 2 illustrates the mechanism behind this probing. After short local adaptation, each client model is evaluated on the same probing inputs at the server, and the entropy of its predictive distribution summarizes how concentrated or uncertain its probing-set predictions are. A more formal bridge is given in Appendix B-F, Proposition 1: under a standard symmetric label-noise transition model, and assuming the probing samples are sufficiently separable and the adapted client predictions are reasonably calibrated, the average predictive entropy is monotone increasing in the client noise rate up to a calibration- and separability-dependent relaxation. This is consistent with classical noisy-label transition models [10], [11] and with confidence-based label diagnosis [13], [21]. Consistent with this mechanism, Figure 3 shows that predictive entropy is positively correlated with the client-side label noise rate in our experimental settings. Therefore, predictive entropy is used as an efficient probing-set reliability signal for identifying reliability outliers before training. Its role is not to recover the true label-noise rate exactly, but to provide an observable ranking signal whose remaining mismatch is captured by the residual term below.

Let $\widetilde{H}_i \in [0, 1]$ be the normalized predictive entropy of client $i$ on the probing set, and define the entropy-based set-level probing

$$\widehat{\xi}_H(S) = \sum_{i \in S} \omega_i(S)\widetilde{H}_i, \qquad \omega_i(S) = \frac{n_i}{\sum_{j \in S} n_j}. \tag{14}$$

Let $z_S$ be the probing-predicted class mixture of the retained client set, and let $p_{\mathrm{prob}}$ denote a target-facing reference prior used by the server. Here, `prob` is shorthand for probing. In our implementation, $p_{\mathrm{prob}}$ is specified without using the true labels of the probing samples. For numerical stability, define the smoothed mixture

$$z_S^{\epsilon} = \frac{z_S + \epsilon\mathbf{1}}{1 + K\epsilon}. \tag{15}$$

Rather than assuming that these proxies are exact, we define

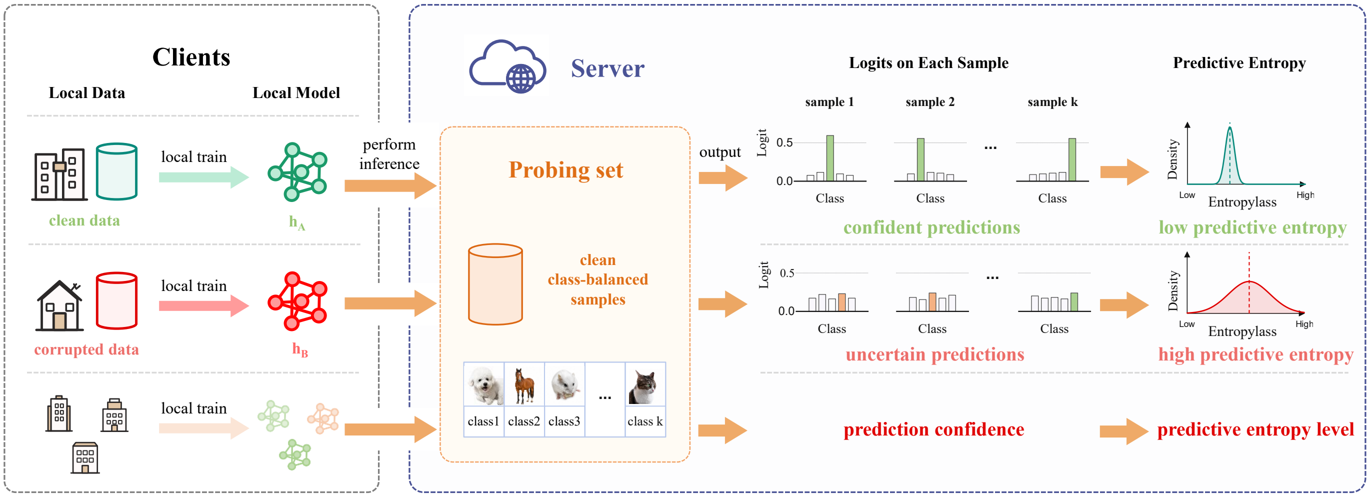


Fig. 2. Illustration of probing-set predictive entropy for client-block reliability estimation. Each candidate client adapts on its private block, and the server scores the adapted model on clean probing inputs without using probing-set labels.

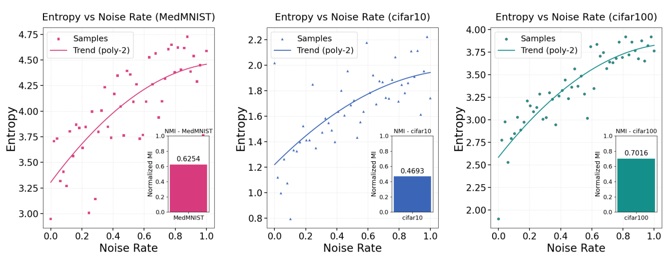


Fig. 3. Empirical relation between predictive entropy and client-side noise in our experiments. The curve shows that entropy increases as the noise level grows, supporting it as a lightweight reliability signal.

the following residual terms:

$$r_H(S;c_H) = \left[\xi_{\text{noise}}(S) - c_H\widehat{\xi}_H(S)\right]_+, \tag{16}$$

$$r_Z(S) = \frac{1}{2}\|\bar{q}_S - z_S\|_1, \tag{17}$$

$$r_T = \frac{1}{2}\|p_{\text{prob}} - q_T\|_1, \tag{18}$$

$$r_\epsilon(S) = \frac{1}{2}\|z_S - z_S^\epsilon\|_1, \tag{19}$$

where $[a]_+ = \max\{a, 0\}$ and $c_H > 0$ is a scaling constant. Here, $r_H$ measures the gap between the population noise-contamination term and the entropy-based probing, $r_Z$ measures the mismatch between the probing-predicted mixture and the clean selected-client prior, $r_T$ measures the mismatch between the server-side reference prior and the target prior, and $r_\epsilon$ accounts for smoothing.

**Theorem 2** (Residualized probing outlier-removal bound)**.** *Under Assumptions 2–4, for any selected subset $S$, any $h \in \mathcal{H}$, and any $c_H > 0$,*

$$\epsilon_T(h) \le \epsilon_S^n(h) + c_H\widehat{\xi}_H(S) + \sqrt{\frac{1}{2}\mathrm{KL}\left(p_{\text{prob}}\|z_S^\epsilon\right)} + R_{\text{prob}}(S;c_H). \tag{20}$$

*where*

$$R_{\text{prob}}(S;c_H) = r_H(S;c_H) + r_Z(S) + r_T + r_\epsilon(S). \tag{21}$$

Theorem 2 separates the observable part of the bound from the residual terms that quantify probing mismatch. Thus, FedOCR optimizes the observable surrogate, while $R_{\text{prob}}(S;c_H)$ captures the remaining gap between probing statistics and population quantities.

**Corollary 1** (probing-consistency outlier-removal surrogate)**.** *Suppose that the probing residual is uniformly bounded over the candidate subsets under consideration, i.e., $R_{\text{prob}}(S;c_H) \le \delta_{\text{prob}}$. Then, for any $\lambda_s > 0$,*

$$\epsilon_T(h) \le \epsilon_S^n(h) + c_H\widehat{\xi}_H(S) + \lambda_s\mathrm{KL}\left(p_{\text{prob}}\|z_S^\epsilon\right) + \delta_{\text{prob}} + \frac{1}{8\lambda_s}. \tag{22}$$

Corollary 1 motivates the computable subset-level outlier-removal surrogate

$$\widehat{\mathcal{B}}(S) = \lambda_m\widehat{\xi}_H(S) + \lambda_s\mathrm{KL}\left(p_{\text{prob}}\|z_S^\epsilon\right), \tag{23}$$

where $\lambda_m$ absorbs the unknown scaling constant $c_H$. This surrogate is the observable part of the residualized probing bound. The residual terms are not directly optimized by FedOCR; they characterize the approximation gap introduced by using probing statistics. Accordingly, the theory provides a probing-consistency justification for FedOCR: the main risk bound specifies which population factors should be controlled, and the residualized bound states when the observable entropy–KL surrogate is aligned with those factors. When the probing residuals are small, minimizing the surrogate controls the target-risk bound up to the residual gap. Additional discussion of the residual terms is provided in Appendix C-A–C-D.

## IV. Method

The theoretical analysis motivates an additive probing objective consisting of an entropy-based predictive unreliability term and a probing distribution-mismatch term. FedOCR implements this objective through the following four components:

- **probing instantiation.** We instantiate the theory-derived probing objective using normalized predictive entropy and probing-KL matching.
- **Client-block scoring.** We convert the additive objective into a local candidate-wise retention score for greedy client-block outlier removal.
- **Noise-adaptive calibration.** We optionally apply noise-adaptive exponents to calibrate the filtering strictness under heterogeneous noise.
- **Variable-size retention.** We use a variable-size prefix rule to determine how many clients should be retained.

Only the first two components are directly tied to the probing objective, while the adaptive exponent and prefix-threshold mechanisms are practical choices for improving robustness under heterogeneous noise.

### *A. Reliability Surrogate via Fixed-Scale Predictive Entropy*

Before outlier removal, each candidate client starts from the current global model and performs a short local adaptation, producing a client model $f_i$. This adaptation is performed once before federated training, using a small fixed local budget $E_{\text{adm}}$, and all candidate clients follow the same protocol. Let $p_i(c \mid x)$ denote the predictive probability of client model $f_i$ on probing input $x \in \mathcal{D}_{\text{prob}}$. We define the average predictive entropy of client $i$ as

$$H_i = -\frac{1}{|\mathcal{D}_{\text{prob}}|} \sum_{x \in \mathcal{D}_{\text{prob}}} \sum_{c=1}^{K} p_i(c \mid x) \log p_i(c \mid x). \tag{24}$$

For a $K$-class categorical distribution, the maximum entropy is $\log K$, achieved by the uniform distribution. We therefore normalize $H_i$ as $\widetilde{H}_i = H_i / \log K$, so that $\widetilde{H}_i \in [0, 1]$ in theory. A value close to 1 indicates near-uniform predictive uncertainty on the probing set, which we use as an observable probing for predictive unreliability. In implementation, $\widetilde{H}_i$ is clipped to $[0, 1]$ only for numerical stability. The fixed-scale reliability score is

$$C_m(i) = \max\{1 - \widetilde{H}_i, \epsilon\}, \tag{25}$$

where $\epsilon > 0$ prevents zero reliability and keeps the negative-log cost finite. At the set level, this score corresponds to the entropy-based probing term $\widehat{\xi}_H(S) = \sum_{i \in S} \omega_i(S) \widetilde{H}_i$, where $\omega_i(S) = n_i / \sum_{j \in S} n_j$. Thus, retaining clients with larger $C_m(i)$ discourages the outlier-removal client set from accumulating high estimated predictive unreliability.

### *B. Representativeness Surrogate via probing-Predicted Class Signatures*

Let $p_{\text{prob}} \in \Delta^{K-1}$ be a server-side target reference prior. For balanced benchmarks, we use the uniform class prior; in deployments, it can be provided by the expected target distribution. FedOCR does not use the true labels of $\mathcal{D}_{\text{prob}}$. For each client $i$, we construct a probing-predicted class signature from hard predictions:

$$r_i(c) = \sum_{x \in \mathcal{D}_{\text{prob}}} \mathbf{1}\left[\arg\max_{c'} p_i(c' \mid x) = c\right], \qquad c = 1, \ldots, K. \tag{26}$$

Let $\hat{r}_i = r_i / (\sum_{c=1}^{K} r_i(c) + \epsilon)$ denote the normalized signature. For a selected subset $S$, we define the probing-predicted mixture as

$$z_S = \sum_{i \in S} \omega_i(S) \hat{r}_i \in \Delta^{K-1}, \qquad \omega_i(S) = \frac{n_i}{\sum_{j \in S} n_j}. \tag{27}$$

In theory, we would like to know the true clean class distribution of the retained client set. However, the server cannot observe each client's clean label prior $q_i$ in FL. Therefore, FedOCR uses the observable prediction signature $\hat{r}_i$ on the probing set as a practical substitute. The resulting mixture $z_S$ should not be viewed as an exact estimate of the clients' true clean label distribution. Instead, it measures which target-relevant classes are covered by the retained client models on the probing set, reflecting the role of label-prior and distribution mismatch in Non-IID generalization [6], [8], [23].

For numerical stability, we apply $\epsilon$-smoothing before computing KL divergence:

$$z_S^{\epsilon} = \frac{z_S + \epsilon \mathbf{1}}{1 + K\epsilon}. \tag{28}$$

For a candidate client $k \notin S$, we measure the distribution mismatch after retaining $k$ by $d(k \mid S) = \mathrm{KL}(p_{\text{prob}} \| z^{\epsilon}_{S \cup \{k\}})$ and convert it into the distribution score

$$C_s(k \mid S) = \exp(-\tau d(k \mid S)). \tag{29}$$

We use the forward direction $\mathrm{KL}(p_{\text{prob}} \| z^{\epsilon}_{S \cup \{k\}})$ because missing reference target classes in the retained pool should be penalized strongly. When $\tau$ is not fixed, we calibrate it from the valid candidate pool $\mathcal{V}$ by setting $\tau = (Q_{q_\tau}(\{d(i \mid \emptyset) : i \in \mathcal{V}\}) + \epsilon)^{-1}$. This calibration makes the score less sensitive to the absolute KL scale of a particular dataset or candidate pool.

### *C. From Additive Surrogate to Local Retention Score*

The additive probing objective in Eq. (23) is a set-level objective. FedOCR does not solve this combinatorial problem exactly. Instead, it greedily minimizes a local candidate-wise approximation, following the common use of greedy procedures in subset and client selection problems [24], [37]–[39]. For a candidate client $k \notin S$, the post-retention KL term $\mathrm{KL}(p_{\text{prob}} \| z^{\epsilon}_{S \cup \{k\}})$ measures the distribution mismatch after retaining $k$, while the entropy term $\widetilde{H}_k$ serves as a local approximation to the marginal increase of the set-level unreliability term.

A direct fixed-score implementation of the additive surrogate is

$$A_0(k \mid S) = \exp\left(-\lambda_s \mathrm{KL}\left(p_{\text{prob}} \| z^{\epsilon}_{S \cup \{k\}}\right) - \lambda_m \widetilde{H}_k\right). \tag{30}$$

This score directly exponentiates the additive surrogate. Since the exponential of a weighted sum factorizes into a product, Eq. (30) naturally motivates a multiplicative score over the two components. Using the bounded monotone scores defined above, namely $C_s(k \mid S)$ in Eq. (29) and $C_m(k)$ in Eq. (25), we obtain the practical retention utility

$$A(k \mid S) = C_s(k \mid S)^{a_{\mathrm{eff}}} C_m(k)^{b_{\mathrm{eff}}}. \tag{31}$$

This form preserves the preference direction of the additive surrogate while keeping both factors bounded and numerically stable. Fixed exponents give the non-adaptive version, whereas adaptive exponents reweight the two components during client-block outlier removal.

The multiplicative form is conservative: a client must be both distributionally useful and sufficiently reliable to obtain a high score. Equivalently, maximizing $A(k \mid S)$ is minimizing the negative-log retention cost

$$\begin{align} R(k \mid S) &= -\log A(k \mid S) \tag{32} \\ &= a_{\mathrm{eff}} \tau \mathrm{KL}\Big(p_{\mathrm{prob}} \| z^{\epsilon}_{S\cup\{k\}}\Big) + b_{\mathrm{eff}}\big[-\log C_m(k)\big]. \tag{33} \end{align}$$

Eq. (32) can be viewed as a local candidate-wise negative-log approximation to Eq. (23), with $-\log C_m(k)$ replacing the marginal entropy cost and the forward KL measuring the post-retention distribution mismatch. For small to moderate entropy values, $-\log(1-\widetilde{H}_k) \approx \widetilde{H}_k$, while highly uncertain clients receive a stronger penalty.

### D. Optional Noise-Adaptive Calibration

The adaptive exponents are practical calibration terms, not a direct consequence of the risk bound. The core theory-aligned version uses fixed weights in Eq. (30). However, a fixed reliability–representativeness trade-off can be suboptimal under heterogeneous noise. When the candidate pool is severely noisy, outlier removal should emphasize reliability; when the pool is moderately noisy, representativeness becomes more important for target-distribution coverage [9], [16].

We estimate the global predictive unreliability level from normalized entropy scores:

$$s = \rho\, \mathrm{Mean}(\{\widetilde{H}_i\}_{i\in\mathcal{V}}) + (1-\rho)\, Q_q(\{\widetilde{H}_i\}_{i\in\mathcal{V}}), \qquad \rho \in [0,1]. \tag{34}$$

Given $s$, define $g = \mathrm{clip}(2(0.5-s), -1, 1)$. Let $a$ and $b$ be base exponents. We use $a_{\mathrm{eff}} = \mathrm{clip}(a(1+\lambda g), a_{\min}, a_{\max})$ and $\bar{b}_{\mathrm{eff}} = \mathrm{clip}(b(1-\lambda g), b_{\min}, b_{\max})$. For datasets with many classes, entropy is harder to calibrate; hence we use the class-scaled exponent $b_{\mathrm{eff}} = \bar{b}_{\mathrm{eff}}(\max\{K/K_b, 1\})^{-\nu_b}$, where $K_b$ is a reference class number and $\nu_b \geq 0$ controls the scaling strength. When $s > 0.5$, $g < 0$, so $a_{\mathrm{eff}}$ decreases and $b_{\mathrm{eff}}$ increases; the outlier-removal rule emphasizes reliability under severe uncertainty. When $s < 0.5$, the rule increases the distribution weight and encourages broader target-distribution coverage. We evaluate the fixed-score variant and the adaptive variant separately in the ablation study.

**Algorithm 1** FedOCR: Fixed-Scale Noise-Adaptive Client-Block Outlier Removal

**Require:** Candidate clients $[N]$, sample sizes $\{n_i\}_{i=1}^N$, probing set $\mathcal{D}_{\mathrm{prob}}$, outlier-removal adaptation budget $E_{\mathrm{adm}}$, greedy horizon $M$, minimum retained budget $K_{\min}$, parameters $\rho, q, a, b, \lambda, \tau, q_\tau, \eta_{\min}, \eta_{\max}, \tau_K$

**Ensure:** Outlier-removal client set $S^\star$

```
V ← [N], S_0 ← ∅, J(S_0) ← 0
for i ∈ V do
    Obtain client model f_i after short local adaptation with budget E_adm
    Compute predictive entropy H_i by Eq. (24)
    Normalize entropy as H̃_i ← H_i / log K
    Compute reliability score C_m(i) by Eq. (25)
    Compute probing-predicted signature r_i by Eq. (26)
end for
Set label-free target reference prior p_prob
Compute global unreliability score s by Eq. (34)
Compute a_eff and b_eff as described in Section IV-D
Calibrate τ if not fixed, and compute adaptive benchmark η(s, K; τ_K) by Eq. (36)
for m = 1, ..., min(M, |V|) do
    for each k ∈ V \ S_{m-1} do
        Form smoothed mixture z^ε_{S_{m-1} ∪ {k}}
        Compute C_s(k | S_{m-1}) by Eq. (29)
        Compute A(k | S_{m-1}) by Eq. (31)
    end for
    k_m ← arg max_{k ∈ V \ S_{m-1}} A(k | S_{m-1})
    S_m ← S_{m-1} ∪ {k_m}
    J(S_m) ← J(S_{m-1}) + log [A(k_m | S_{m-1}) / η(s, K; τ_K)]
end for
S* ← arg max_{S_m : m ≥ K_min} J(S_m)
return S*
```

### E. Variable-Size Greedy Outlier-Removal Set Construction

Many existing client selection methods assume a predefined per-round participation budget, such as a fixed fraction or a fixed number of clients [33], [34], [40]. FedOCR instead lets the final participant pool size be determined by the greedy outlier-removal sequence, and uses a temperature coefficient to regulate the retained cardinality.

We use the full candidate pool $\mathcal{V} = [N]$. FedOCR then constructs a greedy sequence $S_1 \subset S_2 \subset \cdots \subset S_M$, where $M$ is the greedy horizon. At step $m$, the retained client is

$$k_m = \arg\max_{k\in\mathcal{V}\setminus S_{m-1}} A(k \mid S_{m-1}), \qquad S_m = S_{m-1} \cup \{k_m\}. \tag{35}$$

The KL-based objective is not generally submodular, so the greedy rule should be viewed as a practical approximation rather than an approximation-guaranteed optimizer [37], [38]. Its role is to construct a high-scoring retention sequence, from which the final pool size is selected by a noise-adaptive prefix objective.

To decide how many clients to retain after outlier removal, we use a noise-adaptive benchmark with a cardinality temperature coefficient $\tau_K > 0$. Given the global unreliability score $s$, define

$$\eta_0(s) = \mathrm{clip}(\eta_{\min} + (\eta_{\max} - \eta_{\min})s, \epsilon, 1),$$

and

$$\eta(s, K; \tau_K) = \mathrm{clip}\left(\left[\eta_0(s)\big(\max\{K/K_\eta, 1\}\big)^{-\nu_\eta}\right]^{\tau_K}, \epsilon, 1\right). \tag{36}$$

A larger $s$ yields a stricter baseline; $\tau_K$ controls the selection aggressiveness. Because the inner benchmark lies in $(0, 1]$, larger $\tau_K$ decreases the effective retention threshold and tends to retain more clients, while smaller $\tau_K$ yields a more conservative pool.

For each prefix $S_m$, we compute the cumulative threshold-normalized objective

$$J(S_m) = \sum_{t=1}^{m} \log \frac{A(k_t \mid S_{t-1})}{\eta(s, K; \tau_K)}. \tag{37}$$

Equivalently,

$$J(S_m) = \sum_{t=1}^{m} \log A(k_t \mid S_{t-1}) - m \log \eta(s, K; \tau_K),$$

which is an accumulated retention utility with a noise-adaptive cardinality penalty. Under a minimum client constraint $K_{\min}$, the final outlier-removal client set is

$$S^\star = \arg\max_{S_m:\, m \ge K_{\min}} J(S_m). \tag{38}$$

This mechanism adapts the retained client count to the observed quality and distributional complementarity of the candidate pool. In high-uncertainty regimes, the retained set can shrink to avoid unreliable clients; when the candidate pool is mostly reliable and complementary, the method can retain more useful clients.

Algorithm 1 summarizes the complete FedOCR procedure.

## V. Experiments

In this section, we first use a controlled oracle experiment to check whether the theoretical bound matches empirical target risk. We then evaluate FedOCR on three benchmark datasets under heterogeneous label noise and Non-IID settings, and we examine its robustness, cost, probing-set sensitivity, retained-pool composition, and ablation behavior.

### A. Oracle Experiment

The oracle experiment is summarized briefly here; Appendix D provides the full setup and the detailed bound analysis. Figure 4 compares the empirical target risk and the dynamic theoretical upper bound under four oracle retention strategies. The most informative pattern is not that one strategy merely gets a lower final number, but that the bound tracks the same ordering as the empirical risk throughout training. The FedOCR ideal strategy, which retains $C1$, $C2$, and $C3$, drives both curves down fastest because it removes the noisiest client while still preserving the class coverage that the target task needs. The clean-only skewed pool $(C1, C2)$ shows the

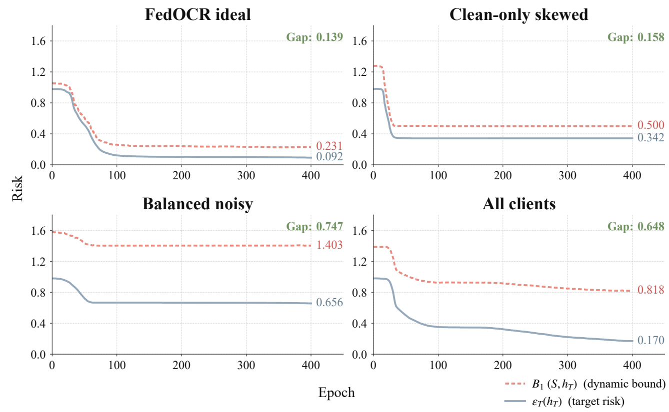


Fig. 4. Empirical target risk and theoretical upper bound during oracle training.

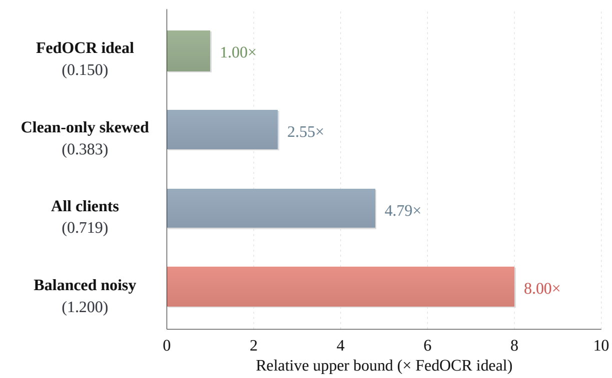


Fig. 5. Theoretical risk bound under different oracle retention strategies.

opposite failure mode: removing the worst noise source helps, but the retained pool is still too narrow, so the risk does not fall as far. The balanced-noisy singleton $C4$ keeps class coverage but leaves the supervision too corrupted to support low risk. Using all clients sits between these extremes because it mixes noisy and mismatched clients in one pool. So the oracle plot does more than rank strategies; it confirms that the theoretical bound captures the same cleanliness-versus-coverage tension that drives the empirical target risk. The bound view tells the same story from another angle. Figure 5 shows that the FedOCR ideal strategy has the tightest bound, the noisy but balanced singleton has the loosest one, and the all-client pool again falls in between. This means the theory is not just abstract: it ranks the oracle retained sets in the same order as the measured target risk.

### B. Main experiments

In this section, we first present a controlled oracle experiment to validate our theoretical analysis. We then evaluate FedOCR on three benchmark datasets under heterogeneous label-noise and Non-IID settings, comparing it with representative noisy-label FL and client-selection baselines, and further studying its efficiency, sensitivity, and ablation behavior.

Table I
Test accuracy (%) of representative noisy-label FL and client-selection methods under heterogeneous label noise on MedMNIST [41], CIFAR-10 [42], and CIFAR-100 [42]. Results are collected at $\alpha = 0.5$ with 50 candidate clients and reported as mean±standard deviation. FedOCR uses the proposed automatic client-block outlier-removal rule in this table. Best results are highlighted in bold and second-best results are underlined.

| Method | MedMNIST | | CIFAR-10 | | CIFAR-100 | |
|---|---|---|---|---|---|---|
| | Linear | Gaussian | Linear | Gaussian | Linear | Gaussian |
| All | 60.89±1.89 | 85.53±0.73 | 39.16±0.31 | 54.38±0.37 | 24.11±0.13 | 31.21±0.23 |
| *Full-round robust training baselines* | | | | | | |
| RFA [43] | 63.38±3.24 | 83.05±0.67 | 35.52±0.10 | 47.68±0.45 | 20.52±0.07 | 30.31±0.23 |
| FedGreed [44] | 83.09±3.17 | 82.22±3.67 | 55.05±0.85 | 54.24±1.68 | 20.79±1.05 | 32.18±0.86 |
| LASA [45] | 54.97±0.51 | 87.52±0.57 | 42.97±0.23 | 56.20±0.31 | 23.69±0.15 | 31.03±0.12 |
| FedNed [46] | 64.29±0.54 | 86.52±0.58 | 39.72±0.09 | 55.41±0.30 | 23.63±0.04 | 32.73±0.11 |
| FedNoRo [16] | <u>85.48±0.41</u> | 87.22±0.42 | 51.14±0.21 | <u>61.09±0.34</u> | 28.56±0.09 | 33.28±0.16 |
| *Training-stage correction baselines* | | | | | | |
| FedCorr [15] | 55.54±13.10 | 85.18±1.02 | 53.39±0.33 | 49.59±0.77 | 26.19±0.17 | 30.67±0.28 |
| FedCS [18] | 56.60±3.02 | 79.12±0.76 | 31.77±0.48 | 47.43±0.25 | 19.84±0.33 | 23.46±0.18 |
| FedFixer [20] | 61.06±1.92 | 85.70±0.76 | 33.53±0.39 | 54.61±0.42 | 23.45±0.17 | 32.71±0.25 |
| FedDiv [17] | 60.75±1.91 | 85.39±0.75 | 41.84±0.74 | 54.49±0.16 | 22.53±0.51 | 31.40±0.12 |
| *Client-selection baselines* | | | | | | |
| PoC [34] | 54.34±2.75 | 79.68±2.12 | 14.79±0.29 | 44.58±0.65 | 25.46±0.21 | 30.05±0.25 |
| GS [47] | 65.50±3.71 | 82.97±0.90 | 46.88±0.45 | 42.43±0.73 | 26.95±0.14 | 23.60±0.23 |
| FedCor [25] | 70.43±3.77 | 86.15±1.32 | 38.22±2.06 | 52.17±0.58 | 26.60±0.16 | 33.41±0.11 |
| FedOCR (ours) | 81.05±0.20 | 85.71±0.37 | <u>56.60±0.24</u> | 58.68±0.41 | 28.73±0.35 | 33.63±0.34 |
| FedOCR + FedCor [25] | 67.82±5.69 | 82.12±0.29 | 56.09±0.18 | 60.62±0.46 | **30.94 ± 0.23** | 32.28±0.17 |
| FedOCR + FedCorr [15] | 79.73±2.15 | <u>88.89±1.11</u> | **57.01 ± 0.84** | 53.07±0.78 | 28.70±0.28 | <u>34.94±0.33</u> |
| FedOCR + FedNoRo [16] | **86.19 ± 0.41** | **91.32 ± 0.39** | 55.68±0.36 | **67.89 ± 0.35** | <u>30.27±0.16</u> | **35.49 ± 0.13** |

Table II
Cost comparison at matched target accuracy. For each scenario, the reference method is selected as a representative strong baseline from the three baseline families considered in Table I, including full-round robust training baselines, training-stage correction baselines, and client-selection baselines. Computation is reported in $10^6$ FEP and communication in GB. Saving denotes the computation / communication reduction of FedOCR-based methods relative to the corresponding reference method. Negative values indicate higher cost than the reference method. Best and second-best costs within each scenario are highlighted in bold and underlined, respectively.

| Scenario | Method | Target | Comp. | Comm. | Saving |
|---|---|---|---|---|---|
| MedMNIST Linear | FedNoRo | 80% | 2.62 | 41.66 | – |
| | FedOCR (ours) | | <u>1.04</u> | **14.66** | <u>60.3%</u> / **64.8%** |
| | FedOCR (ours) + FedNoRo | | **0.84** | <u>18.50</u> | **67.9%** / <u>55.6%</u> |
| MedMNIST Gaussian | FedNoRo | 85% | 3.21 | 54.16 | – |
| | FedOCR (ours) | | **1.78** | **26.16** | **44.5%** / **51.7%** |
| | FedOCR (ours) + FedNoRo | | <u>2.15</u> | <u>35.49</u> | <u>33.0%</u> / <u>34.5%</u> |
| CIFAR-10 Linear | FedCorr | 50% | 13.16 | 49.16 | – |
| | FedOCR (ours) | | **3.08** | **11.17** | **76.6%** / **77.3%** |
| | FedOCR (ours) + FedCor | | <u>3.72</u> | <u>20.75</u> | <u>71.7%</u> / <u>57.8%</u> |
| CIFAR-10 Gaussian | FedCorr | 50% | <u>7.08</u> | <u>34.25</u> | – |
| | FedOCR (ours) | | **4.47** | **18.33** | **36.9%** / **46.5%** |
| | FedOCR (ours) + FedCor | | 7.37 | 54.24 | -4.1% / -58.4% |
| CIFAR-100 Linear | FedCorr | 28% | <u>13.14</u> | 85.30 | – |
| | FedOCR (ours) | | **9.84** | **71.81** | **25.1%** / **15.8%** |
| | FedOCR (ours) + FedCorr | | 16.78 | <u>72.26</u> | -27.7% / <u>15.3%</u> |
| CIFAR-100 Gaussian | FedCorr | 30% | 11.67 | 45.18 | – |
| | FedOCR (ours) | | **8.12** | <u>44.76</u> | **30.4%** / <u>0.9%</u> |
| | FedOCR (ours) + FedCorr | | <u>9.03</u> | **41.89** | <u>22.6%</u> / **7.3%** |

*a) **Experimental setup:*** We evaluate on MedMNIST [41] and CIFAR-10/100 [42]. Unless otherwise specified, the training data are split over $N = 50$ clients and $K = 20$ clients participate in each federated training run. We use Dirichlet partitioning with concentration parameter $\alpha = 0.5$ and a step-imbalance ratio of 5 to simulate Non-IID data [6]–[8]. Label noise is applied symmetrically at the client level, following standard noisy-label robustness protocols [9]–[11]. The main experiments use linear and Gaussian heterogeneous noise to cover two complementary cases: linear noise is a controllable high-noise stress test, while Gaussian noise models a more random corruption pattern that is closer to practical uncertainty. All methods are trained for $T = 200$ global rounds under the same maximum round budget. For FedOCR's admission stage, we use a short 10-epoch local adaptation to compute the filtering indicators, which keeps the pre-screening lightweight and consistent with the deployment-oriented design. For local training, we use Adam with learning rate $1 \times 10^{-4}$ and batch size 64. We compare against three baseline families:

- **Full-round robust training baselines:** RFA [43], FedGreed [44], LASA [45], FedNed [46], and FedNoRo [16].
- **Training-stage correction baselines:** FedCorr [15], FedCS [18], FedFixer [20], and FedDiv [17].
- **Client-selection baselines:** All, RS, EMD, GS [47], PoC [34], and FedCor [25].

Additional implementation details are provided in Appendix E.

*b) **Comparison with baselines**:* We first compare FedOCR with noisy-label FL baselines under linear and Gaussian heterogeneous noise to evaluate noisy-label robustness. Table I reports the accuracy results on MedMNIST [41], CIFAR-10 [42], and CIFAR-100 [42]. Unless otherwise stated, every standalone FedOCR entry in our tables and figures denotes **FedOCR+FedAvg**; we omit "+FedAvg" for brevity because FedAvg is the default downstream optimizer after the admission stage. The table also includes representative client-selection baselines under the same noise settings. The four FedOCR rows separate the admission effect from downstream adaptation: the standalone row applies the proposed pre-training client-block outlier-removal rule and then runs standard FedAvg, while the other rows place the same rule before FedCor, FedCorr, or FedNoRo.

Two patterns stand out. First, standalone FedOCR improves all six dataset-noise settings over full participation, and the largest gains appear under linear noise. This means the admission stage removes the client blocks that do the most damage before their updates spread through training. Second, the combination rows show that FedOCR is a front-end module, not a universal fix. FedOCR+FedNoRo is best in the four settings where a robust downstream learner can use the cleaner pool most effectively, while FedOCR+FedCorr and FedOCR+FedCor win in the two linear cases where the remaining corruption is better matched by correction. So the table supports a simple conclusion: admission gives the main robustness gain, and downstream correction is only a regime-specific add-on.

Table II isolates the system-level picture at matched target accuracy. For each scenario, the reference baseline is a strong non-FedOCR method from Table I that reaches the target. The message is simple: admission is where most wasted work

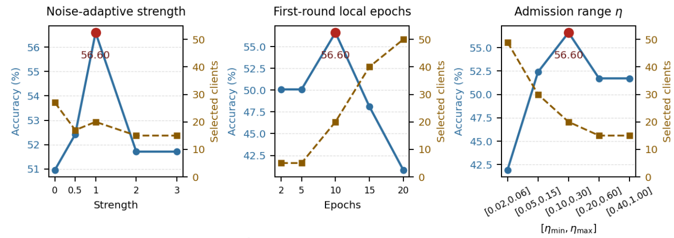


Fig. 6. Hyperparameter sensitivity on CIFAR-10 under linear heterogeneous noise. Each panel reports the test accuracy and the number of retained clients when varying one outlier-removal-stage hyperparameter while keeping the others fixed.

disappears. FedOCR alone reaches the target with the lowest cost in most scenarios because it removes high-risk clients before they consume communication and local epochs. When a downstream correction method is attached, the result becomes a trade-off between extra overhead and better recovery of residual noise. So the table says that the pre-training filter is the main cost saver, and downstream correction helps only when its gain is worth the extra cost.

We also test whether the conclusion changes with the strength of Non-IID partitioning. Table III reports the CIFAR-10 results under linear noise for $\alpha \in \{0.1, 0.5, 1.0\}$. The trend is clear. FedOCR is strongest at the two ends of the range and stays near the top in the middle. That means client-block filtering helps most when skew and corruption are both strong. When heterogeneity is milder, methods that keep more clients can close the gap.

Table III
CIFAR-10 accuracy (%) under linear noise with different Dirichlet Non-IID levels. Results are reported as mean±standard deviation. Best results are highlighted in bold and second-best results are underlined.

| **Method** | $\alpha = 0.1$ | $\alpha = 0.5$ | $\alpha = 1.0$ |
|---|---|---|---|
| All | 26.22±0.21 | 39.16±0.31 | 49.10±0.41 |
| RFA [43] | 23.88±0.34 | 35.52±0.10 | 40.95±0.28 |
| FedGreed [44] | 23.55±1.98 | <u>55.05±0.85</u> | 55.75±1.52 |
| LASA [45] | 26.68±0.28 | 42.97±0.23 | 45.57±0.53 |
| FedNed [46] | 26.01±0.36 | 39.72±0.09 | 49.75±0.32 |
| FedNoRo [16] | <u>29.80±0.41</u> | 51.31±0.42 | <u>56.53±0.20</u> |
| FedCorr [15] | 25.29±1.43 | 53.35±0.33 | 44.93±0.50 |
| FedCS [18] | 20.52±0.37 | 31.77±0.48 | 35.21±1.47 |
| FedFixer [20] | 26.78±0.21 | 33.53±0.39 | 48.89±0.41 |
| FedDiv [17] | 26.47±0.20 | 41.84±0.74 | 49.00±0.38 |
| FedOCR (ours) | $\mathbf{30.27 \pm 0.35}$ | $\mathbf{56.60 \pm 0.24}$ | $\mathbf{57.48 \pm 0.37}$ |

*c) **Hyperparameter sensitivity**:* We study three key hyperparameters in the outlier-removal stage: the noise-adaptive strength, the first-round local epoch budget used to compute filtering indicators, and the threshold range $\eta$. Figure 6 reports the sensitivity results on the CIFAR-10 linear-noise setting. The figure shows a real trade-off. Too little filtering leaves noisy clients in the pool. Too much filtering removes useful coverage. FedOCR works best in the middle, which is what we want from an admission rule.

Table IV
FedOCR probing-set sensitivity on CIFAR-10 [42] under linear heterogeneous noise. Results are reported as the last five rounds' mean±standard deviation, together with the final retained client count and the number of sample scans used by the admission stage.

| Family | Setting | Acc. (%) | Retained | Sample scans |
|---|---|---|---|---|
| Size | 1% | 56.77±0.44 | 14 | 100 |
| | 2% | 56.82±0.43 | 14 | 200 |
| | 5% | $\mathbf{57.10 \pm 0.33}$ | 15 | 500 |
| | 10%(default) | 56.60±0.24 | 15 | 1000 |
| | 20% | 55.04±0.34 | 14 | 2000 |
| | 50% | 55.07±0.45 | 14 | 5000 |
| | 100% | 54.58±0.78 | 14 | 9000 |
| Dirichlet | $IID$(default) | $\mathbf{56.60 \pm 0.24}$ | 15 | 1000 |
| | $\alpha = 10$ | 55.88±0.39 | 15 | 1000 |
| | $\alpha = 1$ | 55.85±0.59 | 13 | 1000 |
| | $\alpha = 0.5$ | 55.25±0.38 | 8 | 1000 |
| | $\alpha = 0.1$ | 56.04±0.34 | 11 | 1000 |
| Noise | clean(default) | 56.60±0.24 | 15 | 1000 |
| | 10% | 56.60±0.24 | 15 | 1000 |
| | 20% | 56.60±0.24 | 15 | 1000 |
| | 30% | 56.60±0.24 | 15 | 1000 |
| | 50% | 56.60±0.24 | 15 | 1000 |

For the strength coefficient, the best result is obtained at strength $= 1$; smaller values keep too many risky clients, while larger values over-prune the pool. For the first-round epoch budget, 10 epochs give the most useful filtering indicators: too few epochs produce weak entropy estimates, while too many epochs make even risky clients look too confident on the probing set and shrink the gap between good and bad clients. For $\eta$, the default range $[0.10, 0.30]$ selects 20 clients and achieves the best accuracy; smaller ranges keep too many noisy clients, while larger ranges are too strict. Additional client-pool-size curves are provided in Appendix Fig. 12.

*d)* ***Probing-set sensitivity****:* This table asks one question: how hard is it to build a probing set for FedOCR? Table IV reports probing-set configurations on CIFAR-10 [42] under the same linear-noise setting used in the dedicated probe study (*ResNet18*, $\alpha = 0.5$, and $\eta \in [0.1, 0.3]$). All runs use a fixed random seed, so identical rows mean that FedOCR selected the same client set and followed the same training trajectory. The size sweep gives the main answer: even a 1% probing set, or 100 images in total, stays very close to the best result. So FedOCR does not need a large clean probe, which keeps the collection cost low. The Dirichlet sweep shows a smaller effect: mild input-distribution changes perturb accuracy only slightly, so the probe does not need to be perfectly IID. The label-noise sweep is flat because FedOCR never uses probing labels. That means label quality is not a real requirement here, which makes the probing set much easier to build in practice.

### C. Analysis of Selected Client Subsets

To understand why FedOCR improves robustness, we analyze the class-wise sample mass and noise rate of the retained client sets. Figure 7 shows the retained pools under linear heterogeneous noise. FedOCR keeps the cleanest pool among

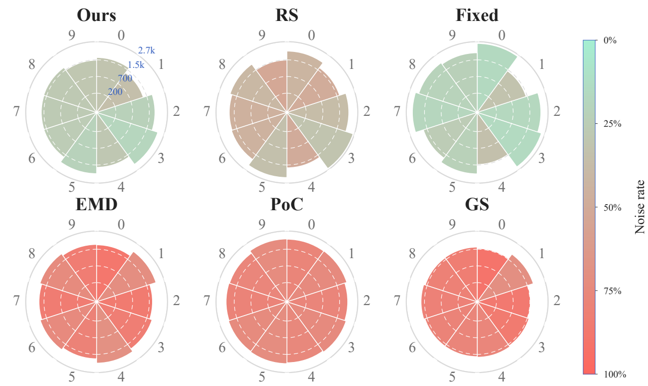


Fig. 7. Class-wise sample-mass and noise-rate profile of the retained client sets under linear heterogeneous noise. The polar radius shows the retained sample mass, and the color encodes the class-wise noise rate from low (green) to high (red).

the compared methods and still spreads mass across classes. RS and Fixed keep broader pools, but they also keep more noisy mass. EMD, PoC, and GS do not solve this trade-off either: they can improve coverage, but they still retain many noisy classes. So the figure says the same thing as the accuracy table: FedOCR keeps clients that are both cleaner and more balanced.

### D. Ablation Study

We ablate the two scoring terms and the adaptive calibration mechanism. Table V clarifies the role hierarchy inside FedOCR. Reliability filtering is the main driver. Once high-risk clients are removed, the retained pool is much easier to optimize, even without any representativeness term. Representativeness alone cannot rescue a noisy pool, so the distribution-only variant still performs poorly. Adaptive calibration adds a smaller but consistent gain by avoiding a fixed reliability-versus-coverage balance across all regimes. So the components play different roles: reliability is the gatekeeper, representativeness protects coverage, and calibration adjusts the balance.

Table V
Ablation study on CIFAR-10 under linear heterogeneous noise. The table isolates the effect of reliability scoring, representativeness scoring, and noise-adaptive calibration.

| Variant | Reliab. | Represent. | Adaptive | Test Acc. (%) |
|---|---|---|---|---|
| Random selection | – | – | – | 39.81±0.32 |
| All clients | – | – | – | 39.67±0.33 |
| Reliability only | ✓ | × | × | 54.60±0.30 |
| Represent. only | × | ✓ | × | 30.69±0.52 |
| w/o adaptive calib. | ✓ | ✓ | × | 53.22±0.16 |
| FedOCR | ✓ | ✓ | ✓ | $\mathbf{56.60 \pm 0.25}$ |

## VI. Discussion

FedOCR uses probing-based signals because the server cannot inspect raw client data under FL privacy constraints [5],

[40]. That makes the method practical, but it also means the probing set matters. Predictive entropy is only a proxy for client unreliability, not a direct estimate of the true noise rate [13], [21]. The KL term is also only a coverage proxy relative to a server-side reference prior [22], [23]. So FedOCR is a coarse pre-training filter by design: it removes risky client blocks before training and leaves residual sample-level noise to downstream correction if needed [15]–[17], [20]. If the client population or target distribution changes over time, the probing set may need to be refreshed.

## VII. Conclusion

This paper studies robust federated learning by removing risky client blocks before training starts. FedOCR builds an outlier-removal client set using two signals: probing-set predictive entropy for reliability and KL-based probing matching for representativeness. Our theory explains why these two signals are related to label noise and target mismatch.

The experiments show three clear results. First, FedOCR improves robustness under noisy and Non-IID settings. Second, it reduces computation and communication cost at matched accuracy. Third, it works with downstream noisy-label correction instead of replacing it. Overall, FedOCR shows that robust FL can benefit from filtering bad clients early, not only from correcting updates during training. Future work will study better probing construction, privacy-preserving probing, and online updates for long-running deployments [5], [32].

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## APPENDIX A
## EXISTENCE OF HARMFUL RISK-PRONE CLIENTS

Let $S \subseteq \mathcal{C}$ be a currently retained client-block set and let $j \in \mathcal{C} \setminus S$ be an additional candidate client. Denote $S^+ = S \cup \{j\}$. Suppose the target-risk upper bound for subset $S$ is written as

$$\mathcal{B}(S) = \epsilon_S^n(h_S) + \xi_{\text{noise}}(S) + \mathcal{D}(T||S).$$

Define the data-benefit term of adding client $j$ as

$$G_j(S) = \epsilon_S^n(h_S) - \epsilon_{S+}^n(h_{S+}),$$

and define the risk-increase term as

$$R_j(S) = \left[\xi_{\text{noise}}(S^+) - \xi_{\text{noise}}(S)\right] + \left[\mathcal{D}(T||S^+) - \mathcal{D}(T||S)\right].$$

If

$$R_j(S) > G_j(S),$$

then

$$\mathcal{B}(S^+) > \mathcal{B}(S).$$

That is, retaining client $j$ worsens the target-risk upper bound, which justifies removing such risk-prone client blocks before training.

*Proof.* By definition,

$$\begin{aligned} \mathcal{B}(S^+) - \mathcal{B}(S) &= \epsilon_{S+}^n(h_{S+}) - \epsilon_S^n(h_S) \\ &\quad + \left[\xi_{\text{noise}}(S^+) - \xi_{\text{noise}}(S)\right] \\ &\quad + \left[\mathcal{D}(T||S^+) - \mathcal{D}(T||S)\right]. \\ &= -G_j(S) + R_j(S). \end{aligned}$$

Therefore, if $R_j(S) > G_j(S)$, then $\mathcal{B}(S^+) - \mathcal{B}(S) > 0$, which proves the claim. □

## APPENDIX B
## ADDITIONAL THEORETICAL DETAILS

### A. Proof of Lemma 1

*Proof.* For any distribution $P$ over $(X, Y)$, the risk of $h$ is defined as $\epsilon_P(h) = \mathbb{E}_{(X,Y)\sim P}[\ell(h(X), Y)]$. Since $\mathcal{Y} = \{1, \ldots, K\}$ is discrete, by the law of total expectation,

$$\epsilon_P(h) = \mathbb{E}_X\left[\mathbb{E}_{Y|X}\left[\ell(h(X), Y) \mid X\right]\right] \quad (39)$$

$$= \mathbb{E}_{x\sim P_X}\left[\sum_{c=1}^{K} P(Y = c \mid x)\ell(h(x), c)\right]. \quad (40)$$

Under Assumption 4, $P_S^c(X,Y)$ and $P_S^n(X,\widetilde{Y})$ share the same input marginal $P_S^X$. Applying Eq. (40) to the clean and noisy mixtures gives

$$\epsilon_S^c(h) = \mathbb{E}_{x\sim P_S^X}\left[\sum_{c=1}^{K} P_S^c(Y=c\mid x)\ell(h(x),c)\right], \tag{41}$$

$$\epsilon_S^n(h) = \mathbb{E}_{x\sim P_S^X}\left[\sum_{c=1}^{K} P_S^n(\widetilde{Y}=c\mid x)\ell(h(x),c)\right]. \tag{42}$$

Therefore,

$$\begin{aligned}
&\left|\epsilon_S^c(h)-\epsilon_S^n(h)\right| \qquad (43)\\
&= \left|\mathbb{E}_{x\sim P_S^X}\sum_{c=1}^{K}\Big(P_S^c(Y=c\mid x)-P_S^n(\widetilde{Y}=c\mid x)\Big)\ell(h(x),c)\right|\\
&\le \mathbb{E}_{x\sim P_S^X}\sum_{c=1}^{K}\left|P_S^c(Y=c\mid x)-P_S^n(\widetilde{Y}=c\mid x)\right|\cdot|\ell(h(x),c)|\\
&\le \mathbb{E}_{x\sim P_S^X}\sum_{c=1}^{K}\left|P_S^c(Y=c\mid x)-P_S^n(\widetilde{Y}=c\mid x)\right|\\
&= \xi_{\mathrm{noise}}(S) \qquad (44)
\end{aligned}$$

The first inequality follows from the triangle inequality, and the second follows from the bounded-loss assumption $0\le\ell\le 1$. □

*B. Proof of Lemma 2*

*Proof.* By the standard variational characterization of total variation distance [48], for any measurable function $f$ with $0\le f\le 1$,

$$|\mathbb{E}_P[f]-\mathbb{E}_Q[f]| \le \mathrm{TV}(P,Q).$$

Applying this inequality to $f_h(x,y)=\ell(h(x),y)$ yields

$$\begin{aligned}
&\left|\epsilon_T(h)-\epsilon_S^c(h)\right| && (45)\\
&= \left|\mathbb{E}_{(x,y)\sim P_T}\left[\ell(h(x),y)\right]-\mathbb{E}_{(x,y)\sim P_S^c}\left[\ell(h(x),y)\right]\right| && (46)\\
&\le \mathrm{TV}(P_T,P_S^c). && (47)
\end{aligned}$$

Since the class-conditional distributions are shared,

$$\mathrm{TV}(P_T,P_S^c) = \frac{1}{2}\sum_{y=1}^{K}|q_T(y)-\bar{q}_S(y)| = \frac{1}{2}\|\bar{q}_S-q_T\|_1. \tag{48}$$

Therefore,

$$\epsilon_T(h) \le \epsilon_S^c(h) + \frac{1}{2}\|\bar{q}_S-q_T\|_1.$$

□

*C. Proof of Theorem 2*

*Proof.* By Theorem 1,

$$\epsilon_T(h) \le \epsilon_S^n(h) + \xi_{\mathrm{noise}}(S) + \frac{1}{2}\|\bar{q}_S-q_T\|_1. \tag{49}$$

By the definition of $r_H(S;c_H)$, we have

$$\xi_{\mathrm{noise}}(S) \le c_H\widehat{\xi}_H(S) + r_H(S;c_H). \tag{50}$$

For the distribution mismatch term, the triangle inequality gives

$$\begin{aligned}
\frac{1}{2}\|\bar{q}_S-q_T\|_1 &\le \frac{1}{2}\|\bar{q}_S-z_S\|_1 + \frac{1}{2}\|z_S-z_S^{\epsilon}\|_1 && (51)\\
&\quad + \frac{1}{2}\|z_S^{\epsilon}-p_{\mathrm{prob}}\|_1 + \frac{1}{2}\|p_{\mathrm{prob}}-q_T\|_1 && (52)\\
&= r_Z(S) + r_\epsilon(S) + \mathrm{TV}(z_S^{\epsilon},p_{\mathrm{prob}}) + r_T. && (53)
\end{aligned}$$

By Pinsker's inequality, also known as the Csiszár–Kullback–Pinsker inequality [49], [50],

$$\mathrm{TV}(z_S^{\epsilon},p_{\mathrm{prob}}) \le \sqrt{\frac{1}{2}\mathrm{KL}\left(p_{\mathrm{prob}}\|z_S^{\epsilon}\right)}. \tag{54}$$

Combining the above inequalities gives Eq. (20). □

*D. Proof of Corollary 1*

*Proof.* By Theorem 2,

$$\begin{aligned}
\epsilon_T(h) &\le \epsilon_S^n(h) + c_H\widehat{\xi}_H(S) && (55)\\
&\quad + \sqrt{\frac{1}{2}\mathrm{KL}\left(p_{\mathrm{prob}}\|z_S^{\epsilon}\right)} + R_{\mathrm{prob}}(S;c_H). && (56)
\end{aligned}$$

For any $x\ge 0$ and $\lambda_s>0$, Young's inequality gives $\sqrt{x/2}\le \lambda_s x + 1/(8\lambda_s)$. Applying this inequality to $x=\mathrm{KL}(p_{\mathrm{prob}}\|z_S^{\epsilon})$ and using $R_{\mathrm{prob}}(S;c_H)\le\delta_{\mathrm{prob}}$ gives Eq. (22). □

*E. Derivation of the Label-Prior TV Distance*

This appendix proves that, under the shared class-conditional assumption, the total variation distance between the target distribution and the selected clean client mixture reduces to the $\ell_1$ distance between their label priors.

Recall Assumption 3. The target distribution and the selected clean client mixture are written as

$$P_T(x,y) = p(x\mid y)q_T(y), \qquad P_S^c(x,y) = p(x\mid y)\bar{q}_S(y),$$

where

$$\bar{q}_S = \sum_{i\in S} w_i(S)q_i.$$

Using the definition of total variation distance,

$$\begin{aligned}
&\mathrm{TV}(P_T,P_S^c) && (57)\\
&= \frac{1}{2}\sum_{y=1}^{K}\int|P_T(x,y)-P_S^c(x,y)|\,dx && (58)\\
&= \frac{1}{2}\sum_{y=1}^{K}\int|p(x\mid y)q_T(y)-p(x\mid y)\bar{q}_S(y)|\,dx && (59)\\
&= \frac{1}{2}\sum_{y=1}^{K}\int p(x\mid y)\,|q_T(y)-\bar{q}_S(y)|\,dx && (60)\\
&= \frac{1}{2}\sum_{y=1}^{K}|q_T(y)-\bar{q}_S(y)|\int p(x\mid y)dx. && (61)
\end{aligned}$$

Since $p(x \mid y)$ is a conditional distribution, $\int p(x \mid y)dx = 1$ for every class $y$. Therefore,

$$\mathrm{TV}(P_T, P_S^c) = \frac{1}{2}\sum_{y=1}^{K} |q_T(y) - \bar{q}_S(y)| = \frac{1}{2}\|\bar{q}_S - q_T\|_1. \quad (62)$$

Thus, under shared class-conditional distributions, the source–target discrepancy is fully determined by the mismatch between the selected client label prior $\bar{q}_S$ and the target label prior $q_T$.

*F. Entropy as a probing for client noise*

We now formalize why average predictive entropy is a useful reliability signal.

**Proposition 1** (Entropy–noise bridge)**.** *Fix client $i$ and assume $K$-class symmetric label noise with rate $\rho_i \in [0, 1 - 1/K)$. Let*

$$T_{\rho_i} = (1-\rho_i)I + \frac{\rho_i}{K-1}(\mathbf{1}\mathbf{1}^\top - I) \quad (63)$$

*be the noise transition matrix. For a probing input $x$, denote the clean posterior by $\eta_i(x) \in \Delta^{K-1}$ and the noisy posterior by*

$$\tilde{\eta}_i(x) = T_{\rho_i}\eta_i(x). \quad (64)$$

*Assume the probing samples are $\delta_{\mathrm{sep}}$-separable in the sense that*

$$\max_c \eta_i(c \mid x) \geq 1 - \delta_{\mathrm{sep}}, \qquad \forall x \in \mathcal{D}_{\mathrm{prob}}, \quad (65)$$

*for some $\delta_{\mathrm{sep}} \in [0, 1)$. Assume further that the local model predictions satisfy*

$$\mathbb{E}_{x\sim P_{\mathrm{prob}}}\big[\mathrm{KL}(\tilde{\eta}_i(x) \,\|\, p_i(\cdot \mid x))\big] \leq \varepsilon_{\mathrm{cal}}. \quad (66)$$

*Define*

$$h_K(\rho) := -(1-\rho)\log(1-\rho) - \rho\log\frac{\rho}{K-1}. \quad (67)$$

*Then the average predictive entropy*

$$\bar{H}_i := \mathbb{E}_{x\sim P_{\mathrm{prob}}}\big[H(p_i(\cdot \mid x))\big] \quad (68)$$

*obeys*

$$\bar{H}_i \geq (1-\delta_{\mathrm{sep}})h_K(\rho_i) - \phi_K\left(\sqrt{\varepsilon_{\mathrm{cal}}/2}\right), \quad (69)$$

*where*

$$\phi_K(\delta) := \delta\log(K-1) + h_2(\delta), \quad (70)$$
$$h_2(\delta) := -\delta\log\delta - (1-\delta)\log(1-\delta). \quad (71)$$

*Consequently, $\bar{H}_i$ is monotone increasing in $\rho_i$ up to a calibration- and separability-dependent relaxation.*

*Proof.* Fix a probing point $x$ and let $c^*(x) := \arg\max_c \eta_i(c \mid x)$. By separability, there exists a distribution $r_x \in \Delta^{K-1}$ such that

$$\eta_i(x) = (1-\delta_x)e_{c^*(x)} + \delta_x r_x, \qquad 0 \leq \delta_x \leq \delta_{\mathrm{sep}}. \quad (72)$$

Applying the symmetric transition matrix gives

$$\tilde{\eta}_i(x) = (1-\delta_x)v_{\rho_i,c^*(x)} + \delta_x T_{\rho_i} r_x, \quad (73)$$

where

$$v_{\rho_i,c^*} := (1-\rho_i)e_{c^*} + \frac{\rho_i}{K-1}(\mathbf{1} - e_{c^*}). \quad (74)$$

By concavity of entropy,

$$H(\tilde{\eta}_i(x)) \geq (1-\delta_x)H\big(v_{\rho_i,c^*(x)}\big) + \delta_x H\big(T_{\rho_i} r_x\big) \quad (75)$$
$$\geq (1-\delta_x)h_K(\rho_i) \geq (1-\delta_{\mathrm{sep}})h_K(\rho_i). \quad (76)$$

Now compare $p_i(\cdot \mid x)$ and $\tilde{\eta}_i(x)$. By Pinsker's inequality,

$$\|p_i(\cdot \mid x) - \tilde{\eta}_i(x)\|_1 \leq \sqrt{2\,\mathrm{KL}(\tilde{\eta}_i(x) \,\|\, p_i(\cdot \mid x))}. \quad (77)$$

The Fannes–Audenaert continuity bound for entropy implies [51]

$$\big|H(p_i(\cdot \mid x)) - H(\tilde{\eta}_i(x))\big| \leq \phi_K\big(\tfrac{1}{2}\|p_i(\cdot \mid x) - \tilde{\eta}_i(x)\|_1\big). \quad (78)$$

Using Pinsker and Jensen,

$$\mathbb{E}_x\big[H(p_i(\cdot \mid x))\big] \geq \mathbb{E}_x\big[H(\tilde{\eta}_i(x))\big] - \phi_K\left(\sqrt{\varepsilon_{\mathrm{cal}}/2}\right). \quad (79)$$

Combining with the lower bound on $H(\tilde{\eta}_i(x))$ yields (69). Finally, $h_K(\rho)$ is strictly increasing on $[0, 1-1/K)$, so the ranking induced by $\bar{H}_i$ agrees with the ranking induced by $\rho_i$ up to the stated relaxation. □ □

*a) **A useful corollary.**:* Since $h_K$ is invertible on $[0, 1 - 1/K)$, (69) implies the conservative upper bound

$$\rho_i \leq h_K^{-1}\left(\frac{\bar{H}_i + \phi_K(\sqrt{\varepsilon_{\mathrm{cal}}/2})}{1-\delta_{\mathrm{sep}}}\right). \quad (80)$$

Under symmetric noise, this immediately yields

$$\xi_{\mathrm{noise}}(S) \lesssim 2\sum_{i\in S} w_i(S)\, h_K^{-1}\left(\frac{\bar{H}_i + \phi_K(\sqrt{\varepsilon_{\mathrm{cal}}/2})}{1-\delta_{\mathrm{sep}}}\right), \quad (81)$$

which motivates the reliability score used in the main text.

*G. Combinatorial optimization and a submodular variant*

The main algorithm uses a KL-based target-matching objective because it directly aligns the retained subset with the probing target distribution. This objective is not guaranteed to be submodular. Nevertheless, the connection to classical subset selection can be made explicit via a closely related facility-location surrogate [37]–[39], [52].

Let $\kappa(i,j) \geq 0$ be a similarity between the predictive signatures $z_i$ and $z_j$ defined in the main text. Consider the coverage function

$$G(S) := \sum_{j=1}^{N} \max_{i\in S} \kappa(i,j), \quad (82)$$

and the set objective

$$F_{\mathrm{sub}}(S) := \lambda_r \sum_{i\in S} r_i + \lambda_s G(S), \qquad \lambda_r, \lambda_s \geq 0. \quad (83)$$

The first term rewards reliable clients; the second term rewards representative coverage of the predictive-signature space.

**Proposition 2** (Submodular subset view)**.** *$F_{\mathrm{sub}}$ is monotone submodular. Consequently, maximizing $F_{\mathrm{sub}}(S)$ subject to $|S| \le M$ is NP-hard in general, and the classical greedy algorithm returns a subset $S_{\mathrm{greedy}}$ satisfying*

$$F_{\mathrm{sub}}(S_{\mathrm{greedy}}) \ge \left(1 - \frac{1}{e}\right) \max_{|S| \le M} F_{\mathrm{sub}}(S) \tag{84}$$

*[37], [38].*

*Proof.* The modular term $\sum_{i \in S} r_i$ is monotone submodular because its marginal gain is constant. The facility-location term $G(S)$ is also monotone submodular: adding an element can only increase the pointwise maxima, and the marginal gain decreases as the current subset grows. A nonnegative linear combination of monotone submodular functions remains monotone submodular. The $(1 - 1/e)$ guarantee then follows from the classical theorem of Nemhauser *et al.* [37]. □ □

*a) **Interpretation.**:* This proposition clarifies the conceptual role of FedOCR. The retained participant pool can be viewed as an *outlier-removal client set*: it compresses the candidate population into a smaller block-level subset that preserves representative coverage while filtering out unreliable or outlying clients.

## APPENDIX C
## DISCUSSION ON PROBING-CONSISTENCY RESIDUALS

This appendix discusses when the residual terms in the residualized probing outlier-removal bound are expected to be small. Recall that Theorem 2 separates the observable probing objective from the approximation residual

$$R_{\mathrm{prob}}(S; c_H) = r_H(S; c_H) + r_Z(S) + r_T + r_\epsilon(S),$$

where

$$\begin{aligned} r_H(S; c_H) &= \left[\xi_{\mathrm{noise}}(S) - c_H \widehat{\xi}_H(S)\right]_+, \\ r_Z(S) &= \frac{1}{2}\|\bar{q}_S - z_S\|_1, \\ r_T &= \frac{1}{2}\|p_{\mathrm{prob}} - q_T\|_1, \\ r_\epsilon(S) &= \frac{1}{2}\|z_S - z_S^\epsilon\|_1. \end{aligned}$$

These residuals are not directly optimized by FedOCR. They quantify the gap between the population quantities in Theorem 1 and the probing statistics used by the algorithm. The probing bound should therefore be interpreted conditionally: when these residuals are small, optimizing the observable entropy–KL surrogate is aligned with reducing the target-risk upper bound.

### A. When is the entropy residual $r_H$ small?

The entropy residual

$$r_H(S; c_H) = \left[\xi_{\mathrm{noise}}(S) - c_H \widehat{\xi}_H(S)\right]_+$$

measures whether the entropy-based probing underestimates the population label-noise discrepancy. It is small when probing predictive entropy is positively aligned with client-level predictive unreliability induced by corrupted supervision.

This condition is more plausible when the following factors hold. First, the probing set should be representative of the target decision regions, so that uncertainty on $\mathcal{D}_{\mathrm{prob}}$ reflects the instability of a locally adapted client model on target-relevant inputs. Second, the local adaptation step should be long enough for the client model $f_i$ to reflect the statistical quality of client $i$, but not so long that it overfits corrupted labels with overconfident predictions. Third, the model should be reasonably calibrated on the probing distribution, so that predictive entropy is meaningful as an uncertainty measure [21]. Under these conditions, clients with stronger label corruption tend to produce less stable or less confident predictions on the probing set, leading to larger $\widetilde{H}_i$.

However, predictive entropy is not a noise-specific statistic. A large entropy value can also arise from hard examples, feature shift, model underfitting, class imbalance, or probing-client mismatch. Conversely, a heavily corrupted client may become overconfident after local overfitting, producing low entropy despite poor label quality. For this reason, FedOCR interprets entropy as a probing-set predictive unreliability measure rather than a direct estimator of the true label-noise rate. In our experiments, we empirically examine the relationship between client noise severity and probing predictive entropy to support this probing choice.

### B. When is the coverage residual $r_Z$ small?

The coverage residual

$$r_Z(S) = \frac{1}{2}\|\bar{q}_S - z_S\|_1$$

measures the mismatch between the clean selected-client label prior $\bar{q}_S$ and the probing-predicted mixture $z_S$. This residual is small when the probing-predicted signatures preserve the relative target-facing class coverage of the retained clients, echoing the importance of source–target distribution alignment in domain adaptation and Non-IID FL [6], [8], [22].

This condition does not require $z_S$ to exactly recover each client's private clean label prior. Instead, it requires that the probing predictions of the locally adapted client models provide a useful class-profile signal for how the retained client pool behaves on target-relevant inputs. The condition is more plausible when the probing set covers the target classes, the adapted client models retain class-discriminative information from their local data, and the hard prediction signatures are not dominated by noise or underfitting. The sample-size weighting in $z_S = \sum_{i \in S} \omega_i(S)\hat{r}_i$ also mirrors the mixture prior $\bar{q}_S = \sum_{i \in S} w_i(S) q_i$, which helps align the probing mixture with the population mixture when the signatures are informative.

The residual can be large when client models are poorly adapted, when local label noise severely distorts decision boundaries, or when the probing distribution is far from the regions where client models are informative. It can also be large under strong feature shift, because a client may have a useful local class distribution but produce biased predictions on the probing set. Therefore, $z_S$ should be understood as a target-facing predictive coverage statistic rather than an exact estimator of private clean label priors.

### C. When is the target-prior residual $r_T$ small?

The target-prior residual

$$r_T = \frac{1}{2}\|p_{\text{prob}} - q_T\|_1$$

measures how well the server-side reference prior approximates the target label prior. It is small when this reference prior is specified to match the target task [22], [23]. For example, in a balanced classification benchmark, a uniform reference prior yields a small $r_T$ when the target prior is also balanced. This does not require using the true labels of the probing samples. In practical deployments, this residual can be reduced by choosing the reference prior according to the expected deployment distribution.

This residual can be large if the reference prior is biased, outdated, or mismatched with the target population. In that case, the KL term $\mathrm{KL}(p_{\text{prob}}\|z_S^{\epsilon})$ encourages matching the reference prior rather than the true target prior. Thus, the reliability of the representativeness surrogate depends on both the reference prior and the probing input distribution.

### D. Effect of the smoothing residual $r_\epsilon$

The smoothing residual

$$r_\epsilon(S) = \frac{1}{2}\|z_S - z_S^{\epsilon}\|_1$$

is introduced only for numerical stability when computing the forward KL divergence. Since

$$z_S^{\epsilon}(c) = \frac{z_S(c)+\epsilon}{1+K\epsilon}, \qquad c = 1,\dots,K,$$

we have $z_S^{\epsilon}(c) > 0$ for every class, which avoids zero-support problems in $\mathrm{KL}(p_{\text{prob}}\|z_S^{\epsilon})$. For small $\epsilon$, the perturbation from $z_S$ to $z_S^{\epsilon}$ is small. Indeed,

$$z_S^{\epsilon} - z_S = \frac{\epsilon\mathbf{1} - K\epsilon z_S}{1+K\epsilon},$$

and therefore

$$\|z_S^{\epsilon} - z_S\|_1 \le \frac{\epsilon\|\mathbf{1}\|_1 + K\epsilon\|z_S\|_1}{1+K\epsilon} = \frac{2K\epsilon}{1+K\epsilon}.$$

Thus,

$$r_\epsilon(S) = \frac{1}{2}\|z_S - z_S^{\epsilon}\|_1 \le \frac{K\epsilon}{1+K\epsilon}.$$

When $\epsilon$ is small, this residual is negligible compared with the main probing terms.

### E. Summary

The residualized bound separates two roles. The observable part,

$$c_H\widehat{\xi}_H(S) + \lambda_s \mathrm{KL}(p_{\text{prob}}\|z_S^{\epsilon}),$$

is the quantity optimized by FedOCR. The residual part,

$$R_{\text{prob}}(S; c_H) = r_H(S; c_H) + r_Z(S) + r_T + r_\epsilon(S),$$

measures how well the probing statistics approximate the corresponding population quantities. Therefore, the theoretical role of FedOCR is conditional rather than unconditional: if the probing set is target-relevant and the adapted client models yield informative uncertainty and class-signature statistics, then minimizing the entropy–KL surrogate is aligned with reducing the target-risk upper bound.

Table VI

CONFIGURATION OF THE ORACLE EXPERIMENT. THE SIX CLIENTS DIFFER IN BOTH CLASS-PRIOR DISTRIBUTION AND LABEL-NOISE RATE. CLIENTS C1–C3 ARE LOW-NOISE BUT INDIVIDUALLY CLASS-SKEWED, WHILE THEIR UNION PROVIDES A BALANCED CLASS DISTRIBUTION.

| **Item** | **Setting** |
|---|---|
| Task type | Three-class non-IID classification |
| Feature dimension | 2 |
| Input distribution | Gaussian class-conditionals |
| Class means | $\mu_1 = (-2, 0),\ \mu_2 = (2, 0),\ \mu_3 = (0, 2)$ |
| Covariance | $\sigma = 0.9$ |
| Target prior | $q_T = (1/3, 1/3, 1/3)$ |
| Label noise | Symmetric noise with rate $\rho$ |
| Noise mechanism | Flip to other $K-1$ classes uniformly |
| Model | Two-layer MLP, $2 \to 16 \to 3$ |
| Activation | ReLU |
| Optimizer | Full-batch gradient descent |
| Learning rate | 0.01 |
| Training epochs | 400 epochs, including epoch 0 initialization |
| Initialization | $\mathcal{N}(0, 0.01)$ with fixed random seed |
| Samples per client | 300 |
| Target test set | 3000 clean samples with balanced class prior |

## APPENDIX D
## ORACLE EXPERIMENT DETAILS

This appendix records the controlled oracle validation that is summarized briefly in the main text. We consider a three-class Gaussian classification problem with two-dimensional features, class-balanced target prior, and six candidate clients that differ in both label prior and symmetric label-noise rate. The purpose of this experiment is not to benchmark practical methods, but to check whether the risk factors in Theorem 1 rank retained client-block sets in the expected order. The four retention strategies isolate the effects of no outlier removal, distribution mismatch, label noise, and the oracle FedOCR-style admission rule.

*a) **Oracle results.**:* The oracle bound and target-risk figures below show the same qualitative pattern described in the main text. The FedOCR ideal subset achieves the lowest theoretical bound because it reduces both the label-noise term and the distribution-mismatch term, while the balanced noisy subset has the largest bound. The empirical target-risk curves further confirm that the FedOCR ideal subset attains the lowest target risk and the smallest bound gap.

## Appendix E
## Experimental Setup Details

### A. Main experiment setup

We evaluate on MedMNIST [41] and CIFAR-10/100 [42]. Unless otherwise specified, the training data are split over $N = 50$ clients and $K = 20$ clients participate in each federated training run. We use Dirichlet partitioning with concentration parameter $\alpha = 0.5$ and a step-imbalance ratio of 5 to simulate Non-IID data [6]–[8]. Label noise is applied symmetrically at the client level, following standard noisy-label robustness protocols [9]–[11]. The main experiments use linear and Gaussian heterogeneous noise. All methods are trained for $T = 200$ global rounds under the same maximum round budget. For FedOCR's admission stage, we use a short 10-epoch local adaptation to compute the filtering indicators, which keeps the pre-screening lightweight and consistent with the deployment-oriented design. For local training, we use Adam with learning rate $1 \times 10^{-4}$ and batch size 64.

### B. Noise Modes

We use two heterogeneous noise modes. In *linear noise*, client noise rates are evenly spaced in $[0, 1]$. In *Gaussian noise*, rates are drawn from $\mathcal{N}(0.3, 0.15^2)$ and clipped to $[0, 1]$. Label noise is applied symmetrically within each client.

## Appendix F
## Supplementary Experimental Curves

To avoid excessive repetition in the main text, we present only the CIFAR-10 result under linear heterogeneous noise as the main illustration. This appendix further provides the complete convergence curves for the remaining settings under the same diagnostic setup, in order to supplement the evidence for the general effectiveness of FedOCR. Across the appendix benchmarks, the main pattern is consistent: FedOCR achieves higher final accuracy and more stable convergence when the same client-block outlier-removal rule is applied under heterogeneous client-level label noise. The linear setting is especially diagnostic because it exposes the method to a wide range of client noise rates within the same federation.

## Appendix G
## Additional Client-Scale Experiment

The client-scale experiment complements the main robustness comparisons. It evaluates whether performance changes monotonically with the number of retained clients. The resulting curves show that overly small subsets can lose useful class coverage, while overly large subsets can reintroduce noisy or biased clients. This supports the variable-size outlier-removal design of FedOCR: the goal is not to maximize participation, but to find a retained participant pool that jointly preserves reliability and target-relevant coverage.

### A. Greedy retained-budget validation

The retained-budget curve below is the appendix version of the greedy-set validation that was summarized in the main text before moving the figure out of the introduction-style discussion. The curve is non-monotonic: retaining too few clients leaves the pool under-covered, while retaining too many reintroduces accumulated noise. This is why FedOCR searches for a high-scoring suboptimal retained set rather than maximizing participation blindly.

## Appendix H
## Additional Tables and Compatibility Results

### A. CIFAR-100 Non-IID Robustness

Table VII reports the CIFAR-100 counterpart of the Non-IID robustness experiment under linear heterogeneous noise. Compared with the CIFAR-10 results in the main text, CIFAR-100 is more challenging and produces lower absolute accuracy. FedOCR consistently improves over full participation by 3.28–4.62 percentage points and remains competitive across all three heterogeneity levels, while FedNoRo achieves the strongest final accuracy in this full-round comparison. This result complements the main-table conclusion: the benefit of pre-training client-block outlier removal is visible beyond CIFAR-10, but the relative ranking can change with the dataset difficulty and heterogeneity level.

Table VII
CIFAR-100 accuracy (%) under linear noise with different Dirichlet Non-IID levels. Results are reported as mean±standard deviation. Best results are highlighted in bold and second-best results are underlined.

| **Method** | $\alpha = 0.1$ | $\alpha = 0.5$ | $\alpha = 1.0$ |
|---|---|---|---|
| All | 18.39±0.15 | 24.11±0.15 | 24.17±0.23 |
| RFA [43] | 17.32±0.17 | 20.52±0.05 | 21.31±0.20 |
| FedGreed [44] | 8.73±1.45 | 20.79±1.05 | <u>29.03±0.85</u> |
| LASA [45] | 17.63±0.11 | 23.69±0.17 | 25.44±0.23 |
| FedNed [46] | 17.57±0.26 | 23.63±0.03 | 24.06±0.20 |
| FedNoRo [16] | **22.56±0.20** | **30.20±0.08** | **32.82±3.79** |
| FedCorr [15] | <u>22.03±1.01</u> | 26.19±0.16 | 24.03±0.24 |
| FedCS [18] | 14.07±0.26 | 19.84±0.20 | 21.43±0.80 |
| FedFixer [20] | 18.66±0.14 | 23.45±0.26 | 25.37±0.23 |
| FedDiv [17] | 17.73±0.13 | 22.53±0.30 | 23.88±0.21 |
| FedOCR (ours) | 21.67±0.27 | <u>28.73±0.72</u> | 28.93±0.52 |

### B. MedMNIST Non-IID Robustness

Table VIII reports the MedMNIST counterpart of the Non-IID robustness experiment under linear heterogeneous noise. The results show that the relative ranking is dataset dependent: FedOCR achieves the strongest accuracy at strong Non-IID, while FedNoRo and FedGreed are strongest at moderate and weak Non-IID levels, respectively. This behavior is consistent with the role of the outlier-removal stage. When heterogeneity is strong, filtering noisy and target-mismatched clients can provide a clearer benefit because full participation mixes highly incompatible local objectives. When heterogeneity becomes weaker, the advantage shifts toward methods that exploit more

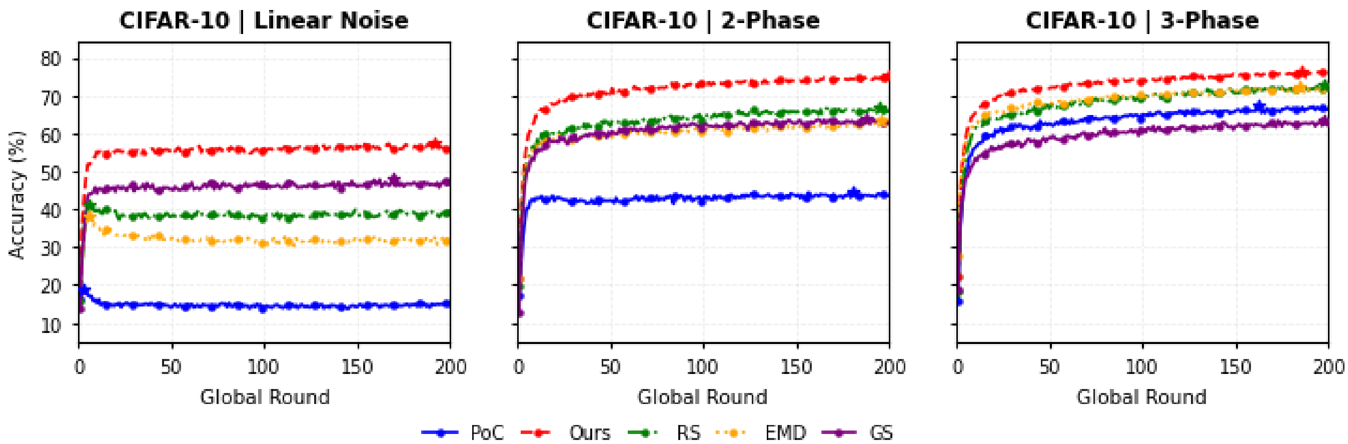


Fig. 8. Test accuracy curves under linear heterogeneous noise in the appendix benchmark suite. FedOCR achieves higher final accuracy and more stable convergence than the compared baselines.

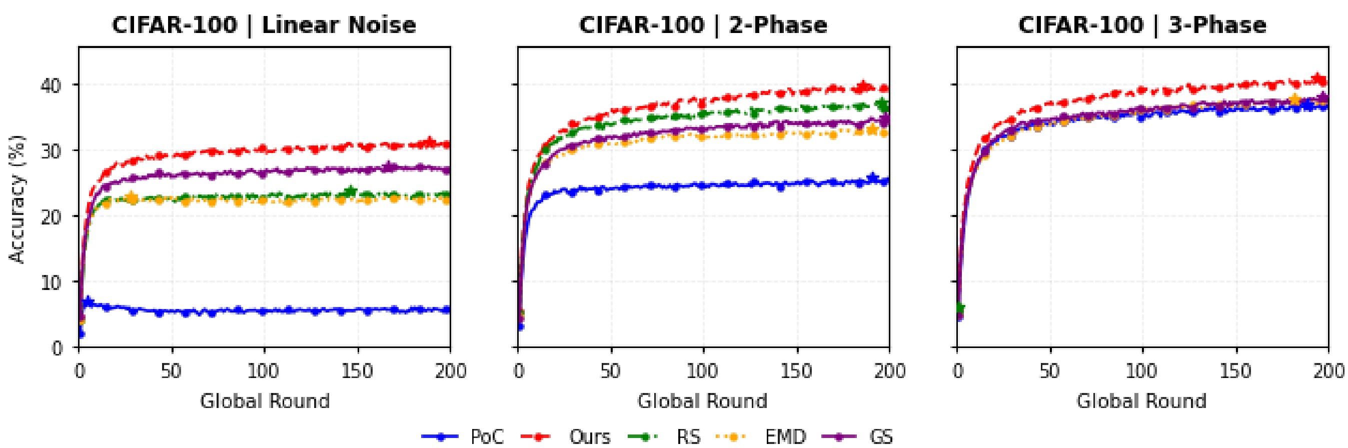


Fig. 9. Additional convergence curves on CIFAR-100 under linear heterogeneous real-imitation noise.


Fig. 10. Additional class-wise clean/noisy sample composition of outlier-removal client sets under linear heterogeneous noise. Positive bars denote clean samples, while negative bars denote noisy samples.

participants during optimization. Therefore, the MedMNIST appendix result should be read together with the cost analysis in the main text: FedOCR is designed to provide a reliable and efficient retained participant pool, rather than to maximize final accuracy in every single dataset column.

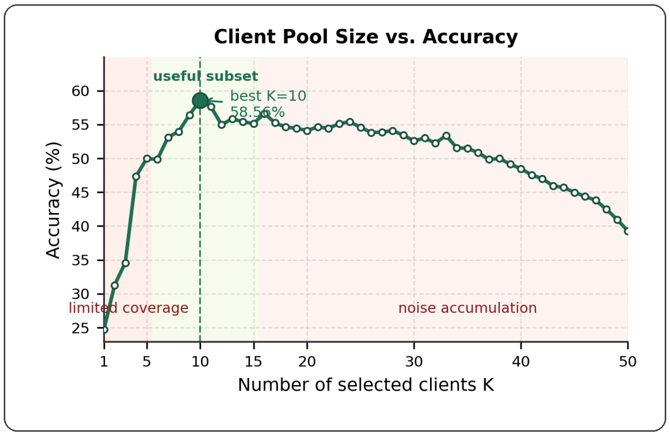


Fig. 11. Final test accuracy under different retained-client budgets in noisy and Non-IID settings. The curve is non-monotonic: admitting too few clients leaves the retained pool under-covered, while admitting too many reintroduces accumulated noise. This motivates our greedy search for a high-scoring suboptimal retained set rather than a blindly maximal one.

Table VIII
MedMNIST accuracy (%) under linear noise with different Dirichlet Non-IID levels. Results use the same best-test-accuracy protocol as Table I and are reported as mean±standard deviation. Best results are highlighted in bold and second-best results are underlined.

| **Method** | $\alpha = 0.1$ | $\alpha = 0.5$ | $\alpha = 1.0$ |
|---|---|---|---|
| All | 46.55±1.44 | 60.89±1.89 | 67.68±1.72 |
| RFA [43] | 38.82±1.50 | 63.38±3.24 | 61.60±1.47 |
| FedGreed [44] | 21.18±0.00 | <u>83.09±3.17</u> | **89.33±2.27** |
| LASA [45] | 38.97±0.86 | 54.97±0.51 | 57.08±1.91 |
| FedNed [46] | 45.50±2.12 | 64.29±0.54 | 67.96±1.49 |
| FedNoRo [16] | 33.92±2.13 | **85.48±0.41** | 78.07±0.51 |
| FedCorr [15] | 4.88±0.86 | 55.54±13.10 | 69.09±9.95 |
| FedCS [18] | 24.66±1.40 | 56.60±3.02 | 79.06±1.39 |
| FedFixer [20] | <u>46.72±1.47</u> | 61.06±1.92 | 67.85±1.75 |
| FedDiv [17] | 46.41±1.46 | 60.75±1.91 | 67.54±1.74 |
| FedOCR (ours) | **47.71±1.08** | 81.05±0.80 | <u>86.82±0.10</u> |

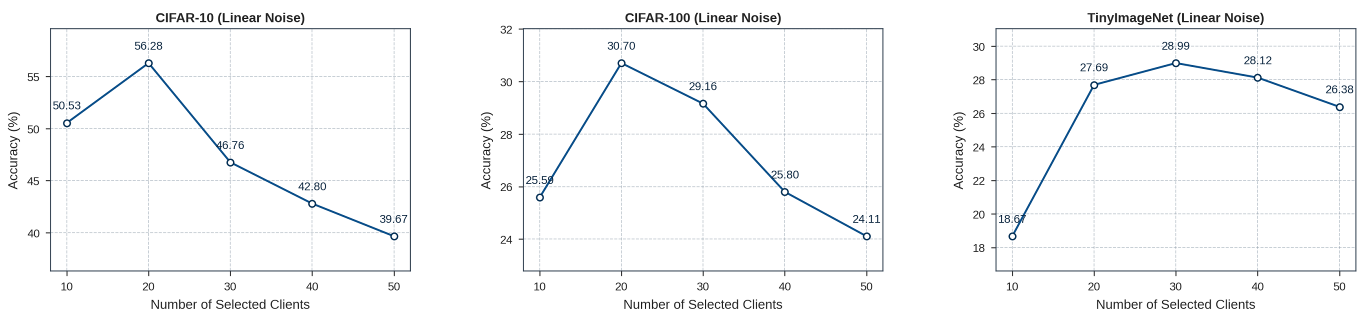


Fig. 12. Effect of the retained client pool size on final test accuracy under coupled label noise and Non-IID heterogeneity. Performance is non-monotonic with respect to the number of participating clients, showing that retaining an appropriate outlier-removal subset can outperform full participation.